CHAPTER 3

Rotation and Single-Particle Motion

3.1 Introduction

The experimental level schemes of nuclei show an enormous complexity. On the way to understanding at least the basic features of their structure, we have introduced in the first part of this book two rather contrasting models: On the one hand the liquid drop model describes collective phenomena, such as vibrations and rotations, where many nucleons are involved. On the other hand the shell model treats the individual nucleons as independent particles and provides an understanding of single-particle excitations. However, these two models are only limiting cases that are never realized exactly in nature. We always have some deviations and we usually find all kinds of transitions between these extreme models.

For a deeper understanding of the underlying structures, we have to solve, in principle, the nuclear many-body problem and investigate in which limits solutions are provided corresponding to the above simple phenomenological pictures, and in which cases new models must be introduced to obtain a simple description. In the following chapters of this book, we will treat such microscopic methods in much detail, and will reproduce many of the results of the phenomenological models. Although such theories are extremely useful for an understanding of principal questions, they usually involve a terrible amount of numerical effort and are not very useful for a fast and qualitative interpretation of experimental data.

Therefore, it seems to be worthwhile going further on the phenomenological path in this chapter and trying to describe more complicated spectra by a combination of the collective model and the single-particle model. In the last chapter, we saw that such a combination is very useful in describing bulk properities such as nuclear binding energies and deformations. In that case, we had to deal with fluctuations of the level density, and we did not have to take into account the individual degrees of freedom in detail. Now we want to describe individual spectra. For this we have to again try a combination of the single-particle and collective models. We can, for instance, couple one or a few particles to a collective rotor or vibrator. This so-called "*unified model*" was introduced by Bohr and Mottelson [Bo 52, BM 53], and has been described in great detail in many textbooks (see, for instance, [Da 68, EG 70, Ro 70, SF 74, or BM 75]). In the case of collective vibrations, we will deduce the corresponding Hamiltonian and discuss the model of a particle coupled to a vibrator in Chapter 9. In the case of rotations, such a derivation from first principles has been missing till now. Methods have been developed to describe rotations microscopically (see Chap. 11), but they turn out to be extremely complicated. Only in the limit of strong deformations is it possible to deduce a simple rotational model, the cranking model.

In the last few years considerable experimental effort has been made to investigate rotations, and a great variety of new phenomena has been observed. Since these phenomena can be understood to a large extent by the interplay between single-particle motion and collective rotational motion, we think it is necessary to introduce in this chapter two theoretical but phenomenological models which deal with these effects.

In Section 3.2 we give a general survey of the features one expects in rotating nuclei. In Section 3.3 we discuss the rotational part of the unified model, the so-called particle-plus-rotor model (PRM), and in Section 3.4 we present the microscopic cranking model.

We want to emphasize, however, that we do not give a complete description of nuclear rotation in this chapter. The residual interactions in the form of pairing correlations play an important role in this context. They will be treated in Chapter 7.

3.2 General Survey

3.2.1 Experimental Observation of High Spin States

New experimental techniques, namely, heavy ion fusion reactions [MG 63, SLD 65], Coulomb excitation with heavy projectiles [HZ 53, AW 66, 74, WCL 76], and pion capture reactions [EAD 75] have made it possible to excite nuclear states with angular momenta large enough to generate major modifications in the nuclear structure (for a review, see [JS 73, St 76, LR 78]).

The most important such reactions are of the (*HI, xn*) type, where one bombards the target with heavy ions (α, Ne, Ar, etc.) carrying a large amount of orbital angular momentum. After fusion, the combined system evaporates some neutrons and ends up in highly excited states with very large spin (depending on the projectile and the incoming energy up to $\sim 100\ \hbar$). This highly excited final nucleus then de-excites in a cascade of γ-radiation. The fastest transitions are *statistical E1-transitions*. They carry away much energy, but only a few units of angular momentum. In a plot of the energy versus the angular momentum (Fig. 3.1), these statistical cascades correspond to nearly vertical lines. They end up at the so-called *yrast line*,* the line which connects the levels of lowest energy to each angular momentum (or the levels with highest angular momentum at a given energy). The rest of the cascade has to follow this line. For deformed nuclei these are *collective E2-transitions* ($\Delta I = 2$), which finally go over into the well-known ground state rotational band (see Sec. 1.5).

Until now, only the last part of these cascades could be resolved into discrete lines (see Fig. 3.5). The highest angular momenta observed in this way lie between 30 and 40 $\hbar$. The rest of the cascade is measured as a γ-continuum, and one can draw only indirect conclusions from these data [GG 67, BSC 75, SBC 76, Di 76, HLM 78, ABH 78, WF 78b]. There are, however, indications that collective transitions along lines parallel to the yrast line (see Fig. 3.1) play an important role, and that there is a

*The Swedish word *yrast* means "fastest rotating" [Gr 67]. One also uses the name *yrare* for the level with the second lowest energy at each I-value.

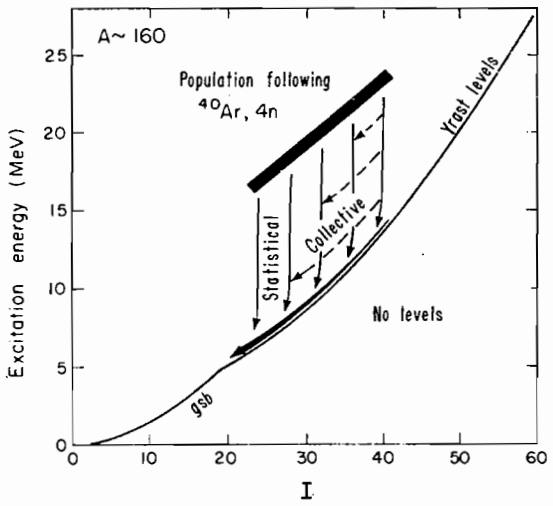


Figure 3.1. Excitation energy as a function of angular momentum I for the reaction (^{40}Ar, 4*n*). After the evaporation of four neutrons, the nucleus decays by a γ-cascade of statistical $E1$- and collective $E2$-transitions down to the yrast line, and then to the ground state by a series of $E2$-transitions along this line. The yrast band shows an increasing intensity, because it collects all the statistical transitions ("side feeding"). (From [SBD 77].)

competition between collective and statistical effects in the region of a few MeV above the yrast line [SBD 77].

3.2.2 The Structure of the Yrast Line

In Section 1.7 we saw that a *classical liquid drop* [CPS 74] rotates at low angular frequencies around the symmetry axis of its oblate shape (Hiskes regime, Fig. 3.2a). Only for very high angular velocities does it undergo a phase transition to the Beringer–Knox regime, where it has a triaxial, but nearly prolate, shape and the rotational axis is perpendicular to the approximate symmetry axis (Fig 3.2b). For still higher frequencies it finally fissions (Fig. 3.2c).

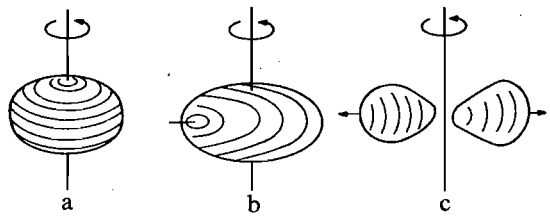


Figure 3.2. The behavior of a classical liquid drop for increasing angular velocity.

The *real nucleus*, however, is a quantum mechanical system. It shows shell effects (see Chap. 2.9), which cause stable deformations already in the ground state for some regions of the periodic table. The yrast line corresponds to the lowest energy for each angular momentum; all the excitation energy of several tens of MeV with respect to the ground state is rotational energy and used to generate angular momentum. Consequently, the level density along this line is low, much lower than the level density of the $I=0$ states at the corresponding energy. It resembles that of the ground state, though we will see that there are characteristic deviations from it. Since the nucleus is cold in the yrast region, we can expect a high degree of order. Shell effects play an important role.

The prolate deformations caused by the shell effects are of the same order of magnitude as the oblate deformations of a classical drop at high angular velocities (see Sec. 1.7). We therefore expect a delicate interplay between macroscopic centrifugal effects and microscopic shell structure when we study the nuclear shapes as a function of the angular momentum.

Let us first study the case of well-deformed heavy nuclei (for instance, in the rare earth region). In the ground state they show a prolate axial symmetric quadrupole deformation caused by shell effects. The levels in the corresponding deformed potential are occupied pairwise by nucleons with the opposite single-particle angular momentum ($\pm\Omega$). We will see in Chapter 6 that the two nucleons in such a pair do not move independently, as was assumed in the last chapter, but are coupled by a pairing force to the so-called Cooper pairs with spin zero; that is, these deformed nuclei

show a *superfluid behavior* and their ground state energy is lowered by a few MeV.

With increasing angular momentum, we can distinguish several regimes where the yrast states show quite a different structure [BM 74]:

(a) For low angular momenta $I=0,2,4,\ldots$, the yrast line follows the *ground state rotational band*, as discussed in Section 1.5. The rotation is collective, that is, it has to be perpendicular to the symmetry axis (see Fig. 3.3a, where we have indicated the coupled pairs of nucleons oriented along the symmetry axis).

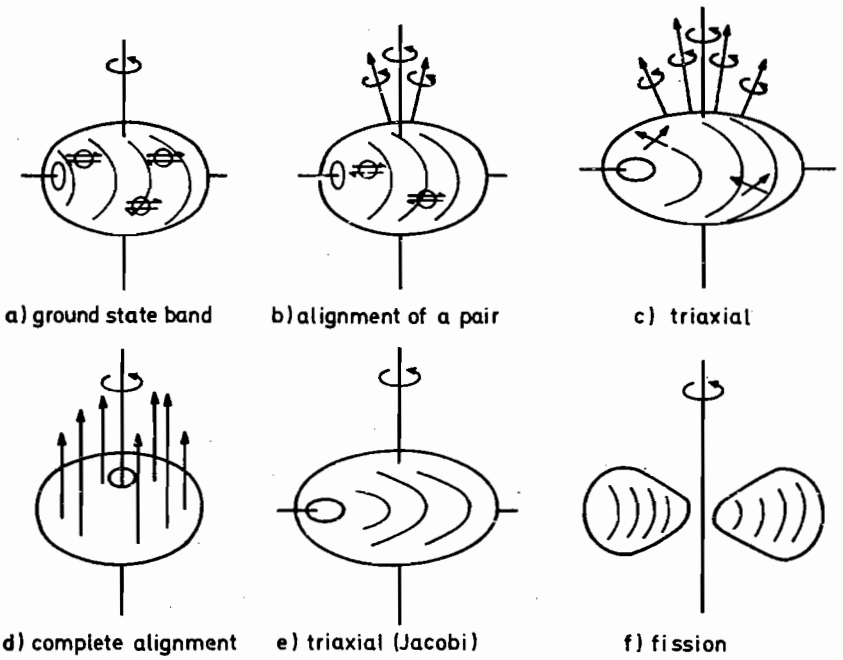


Figure 3.3. Possible structures along the yrast line of a deformed nucleus.

The nucleus feels a slowly rotating deformed potential (for a more detailed discussion, see Sec. 3.4). The Coriolis forces act on both spins of a pair with opposite angular momenta $\pm\Omega$ in opposite directions and try to align them parallel to the rotational axis, that is, they try to break the Cooper pairs [*Coriolis anti-pairing (CAP) effect*]. However, for low angular momenta the Coriolis forces are weak and unable to break pairs: The nucleus rotates more or less with the same structure as the ground state.

We have to emphasize, however, that in a microscopic picture the total angular momentum has to be generated by the angular momenta of single nucleons. In a collective rotation this is achieved by aligning all the particles a little bit in the direction of the rotational axis. For each pair this effect is very small; the spins of the pairs are still oriented nearly anti-parallel along the symmetry axis of the potential. By summing up all the small contributions we obtain the total angular momentum $I=2,4,\ldots,$.

(b) Going to higher and higher angular momenta, the Coriolis force, increases more and more and at a certain angular momentum it will be able to break pairs. Therefore, at some critical angular momentum I_c we expect that all pairs are broken. The pairing correlations break down completely and a *phase transition* from the superfluid phase to the normal phase is observed. This effect was predicted by Mottelson and Valatin in 1960 [MV 60]. More detailed investigations, however, have shown that the Coriolis force is proportional to the size of the single-particle angular momentum j of the nucleon under consideration. We have seen in Section 2.8 that the nucleons in the vicinity of the Fermi surface belong to subshells with rather different j-values, and we expect those nucleons with large j-values to align first along the rotational axis. These are usually the levels with high spin and opposite parity, shifted downwards by the spin orbit term, as for instance the $i_{13/2}$ shell for neutrons in the rare earth region (Stephens–Simon effect [SS 72a]).

In the second regime of the yrast line (Fig. 3.3b), we therefore expect *alignment processes* of one or the other broken pairs, whereas the rest of the nucleus stays more or less unchanged. The yrast band can then no longer be identified with the ground state band, but rather with a band of two aligning particles sitting on the rotating core. Two $i_{13/2}$ particles can contribute 12 units of angular momentum to the rotation. Such an alignment is therefore connected with a rapid increase of the angular momentum I as a function of the collective angular velocity and leads to a series of anomalies in the spectra [JRS 71, JRH 72], for example, the "back-bending phenomenon" (see below).

(c) Each alignment process is connected with a certain change in the collective properties by its influence on the mean field: Broken pairs no longer contribute to the pairing correlations; because of the blocking effect (see Sec. 6.3.4.) these correlations will disappear completely after a few alignments. Particles aligned to the rotational axis have an oblate density distribution, with the rotational axis as the symmetry axis. We therefore get triaxial admixtures to the prolate density distribution of the core.

In a third regime (Fig. 3.3c) which should correspond roughly to angular momenta $30 \lesssim I \lesssim 50$, we expect the Coriolis and centrifugal forces to produce effects comparable in strength to the shell structure effects, namely, changes in the shapes to *triaxial deformations*. Such a system without a symmetry axis shows more collective states than the axial symmetric rotor (see Sec. 1.5.3): We expect a sequence of rotational states parallel to the yrast line corresponding to a *wobbling motion*.

(d) If an essential part of the nucleons are aligned parallel to the rotational axis, we finally expect an *axially symmetric oblate shape*.

The nuclear wave function is symmetric with respect to rotations around this axis. The rotation is no longer collective. Instead, the total angular momentum is made up by reoccupation of distinct particles in the deformed well.

This kind of motion is called "*single-particle" rotation*. The energy of the states along the yrast line in this region is determined by the single-particle energies in the rotating oblate well. The energy differences between adjacent states vary statistically. Only on the average do they follow an $I \cdot (I+1)$ law with the moment of inertia of a rigid rotor (see Sec. 3.4.6.). The transition from one state to the next corresponds to a reoccupation, and the matrix elements should adopt single-particle values, that is, the transition probability should be drastically reduced.

Because of the statistical nature of the levels in that region of the yrast line, one also expects long lived *high spin isomeric states*, so-called *yrast traps* [BM 74] which could eventually produce a delayed decay such that we could observe the whole yrast cascade of discrete lines without the background of all statistical transitions.

(e) For very large angular momenta we finally expect only the macroscopic properties to be essential, that is, the nucleus should again undergo a transition from oblate to triaxial and prolate shapes before it fissions (Fig. 3.3e, f).

So far, only discrete levels in the regimes (a) and (b) have been observed; the rest of these considerations is to a large extent speculation. We will see in Sec. 3.4.5 the kind of models one has used to obtain theoretical predictions. In fact, in the calculations one has found all the different regimes discussed above.

It seems to be possible, however, that many nuclei do not pass through all these stages. In particular, weakly deformed or spherical nuclei adopt from the beginning the regime (d), namely, a rotation of an oblate shape around the symmetry axis before going triaxial and fissioning. On the other hand, in many well-deformed nuclei the change in deformation produced by the alignment processes is not strong enough to compensate the shell effect of a prolate deformation. They only become triaxial (Fig. 3.3c, d), not oblate, before fissioning. Experimentally, one has observed longliving high spin isomers (yrast traps) only for weakly deformed nuclei (see Sec. 3.4.6). This seems to be in agreement with theoretical calculations.

Besides the yrast band, one has observed many other rotational bands in the low and *medium spin region*. They are based on more complex configurations, such as quasi-particle excitations or vibrational states.

Depending on the size of the j shell orbitals, which are involved in such configurations, one can imagine a large variety of alignment processes. For example, in an odd mass nucleus the alignment of the odd neutron in the $i_{13/2}$ shell, which leads to so-called *decoupled bands*. Another example is the rotational band in an odd-odd nucleus, where only the neutron in the $i_{13/2}$

shell aligns, whereas the proton stays oriented along the symmetry axis. Such a band has a *semidecoupled* structure [NVR 75].

The particle-plus-rotor model discussed in Section 3.3 provides a phenomenological method to describe such processes. In that section we will come back to some examples of this type in more detail.

3.2.3 Phenomenological Classification of the Yrast Band

Over the years some phenomenological models have been introduced to characterize the properties of rotational bands. In the low spin region (Fig. 3.3a), the spectra often follow exactly the $I\cdot(I+1)$ law, but for higher I-values there occur deviations to a greater or lesser extent*. In order to describe the deviations from the ideal rotational spectrum, one often applies the following parametrization:

$$E(I)=A\cdot I\cdot(I+1)+B\cdot(I(I+1))^2+C\cdot(I(I+1))^3+\cdots. \tag{3.1}$$

It turns out that in many cases the convergence of this expansion is rather poor and an expansion in the *angular frequency* ω is more appropriate.

In principle, ω is not a measurable quantity. We can define it, however, semiclassically as

$$\omega=\frac{dE}{dJ}, \qquad \text{with} \quad J=\sqrt{I(I+1)}\ . \tag{3.2}$$

Replacing the differential quotient by a quotient of finite differences,† we obtain a definition of an "experimental" value for the *angular velocity*‡

$$\omega_{\text{exp}}\simeq\left.\frac{\Delta E}{\Delta\sqrt{I(I+1)}}\right|_{I,I-2}\simeq\frac{E(I)-E(I-2)}{\sqrt{I(I+1)}-\sqrt{(I-2)(I-1)}}\ . \tag{3.3}$$

The *moment of inertia* is defined by

$$\mathcal{I}=\frac{J}{\omega}=\frac{1}{2}\left(\frac{dE}{dJ^2}\right)^{-1}\simeq\frac{2I-1}{\Delta E_{I,I-2}}\ . \tag{3.4}$$

With these definitions, we can calculate values of ω and $\mathcal{I}$ for each level of the yrast band.

Harris [Ha 65a] proposed the following parametrization of the spectrum:

$$E(I)=\alpha\omega^2+\beta\omega^4+\gamma\omega^6+\cdots. \tag{3.5}$$

Odd powers in ω do not occur, since E cannot change by reversing the angular velocity. For ω as a function of I we can either choose the experimental value (3.3) or avoid the ambiguity of its definition by using a similar expansion of J. From

$$\frac{dE}{d\omega}=\frac{dE}{dJ}\frac{dJ}{d\omega}=\omega\frac{dJ}{d\omega} \tag{3.6}$$

we obtain

$$J(I)=\sqrt{I(I+1)}=2\alpha\omega+\frac{4}{3}\beta\omega^3+\frac{6}{5}\gamma\omega^5+\cdots.$$

* For a compilation of such data, see [SHJ 73, SSM 75].

† This replacement is not unique, however, and some groups use different prescriptions (see [JS 73, So 73, LR 78]).

‡ Within this chapter we always use the units $\hbar=1$.

Of course, this expansion is valid only as long as J is not a multivalued function of ω, that is, it is of no use in the backbending region (see Sec. 3.2.4.).

Another model which is widely used for the classification of rotational and even transitional nuclei is the *variable moment of inertia* (VMI) model [MSB 69, SDG 76]. The moment of inertia $\mathcal{I}$ is considered as a variable on which the intrinsic energy V depends:

$$E(I, \mathcal{I}) = \frac{1}{2\mathcal{I}} I(I+1) + V(\mathcal{I}). \tag{3.7}$$

According to the variation principle (discussed in Sec. 5.2), the total energy E has to be minimized for a fixed value of I with respect to $\mathcal{I}$. This determines the functional dependence $\mathcal{I}(I)$. Assuming we know $V(\mathcal{I})$, $\mathcal{I}(I)$ is implicity given by

$$\left.\frac{dE}{d\mathcal{I}}\right|_I = -\frac{1}{\mathcal{I}} \frac{I(I+1)}{2\mathcal{I}} + \frac{dV}{d\mathcal{I}} = 0.$$

Usually one expands on $V = \frac{1}{2} C(\mathcal{I} - \mathcal{I}_0)^2$, where $\mathcal{I}_0$ is the value of $\mathcal{I}$ at $I = 0$. The coefficients C and $\mathcal{I}_0$ are adjusted to the experimental spectrum $E(I)$. With the identification $J = \omega \cdot \mathcal{I}$ this corresponds to the Harris model to order ω^4, but the ansatz (3.7) is certainly more general [DB 73, SDG 76].

3.2.4 The Backbending Phenomenon

In the region between 10 and 20 units of angular momentum, an anomaly is observed in the yrast band of many nuclei. It can be most easily demonstrated if one plots the moment of inertia $\mathcal{I}$ as a function of ω^2. In lowest order in the VMI model ($\mathcal{I} = \mathcal{I}_0 + b\omega^2 + \cdots$) this should give a straight line. The deviation from a constant is then a measure of the validity of the $I \cdot (I+1)$ law. Figure 3.4 gives two examples for such curves, measured by the Jülich group [LR 78].

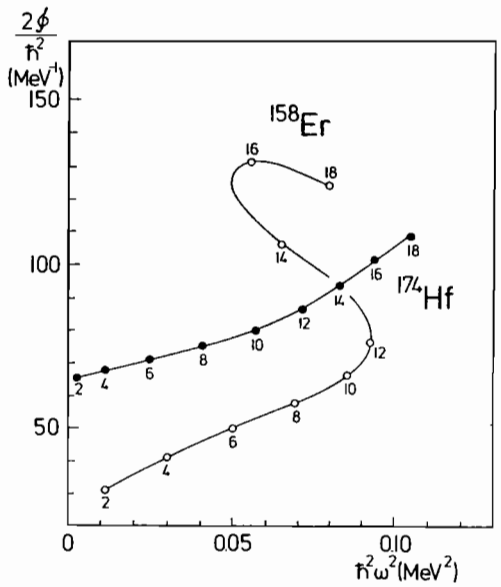


Figure 3.4. Moment of inertia $\mathcal{I}$ as a function of the rotational frequency squared for ^{158}Er and ^{174}Hf.

For low spin values one indeed finds straight lines. In the nucleus ^{174}Hf the deviations are, in fact, smooth. In ^{158}Er, however, a very steep increase occurs for certain I values, the curve even bending backwards, ("back-bending" phenomenon [JRH 72]). This means experimentally that the transition energy $\Delta E_{I,I-2}$, which should increase linearly with I for the constant rotor as

$$\Delta E_{I,I-2} = \frac{1}{2\vartheta}(4I-2), \tag{3.8}$$

does not increase, but decreases for certain I values. Figure 3.5 shows the experimental data that correspond to Figure 3.4.

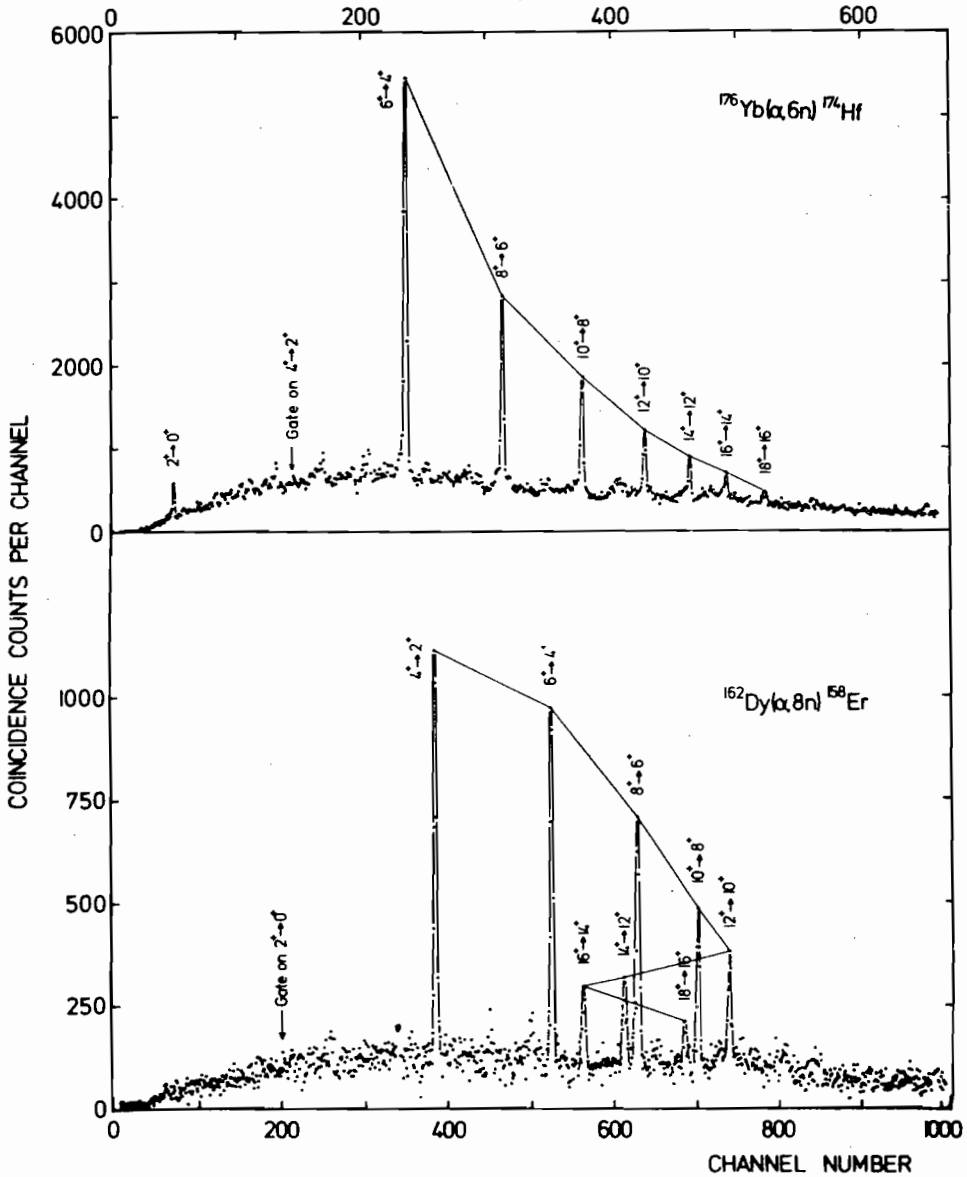


Figure 3.5. The reactions ^{176}Yb $(\alpha, 6n)$ ^{174}Hf and ^{162}Dy $(\alpha, 8n)$ ^{158}Er. The last reaction shows the anomaly called backbending in the I region 10^+-14^+. This is revealed in combining descendent peak heights by straight lines. (From [LR 78].)

Such a phenomenon can easily be reproduced as an effect due to the crossing of two bands (in some cases one has even observed the second band experimentally, see Fig. 7.6 and [KSM 78]). To see this, let us assume that the two bands have different moments of inertia (Fig. 3.6) corresponding to two parabolas in an E versus I plot. If $\mathcal{I}_2 > \mathcal{I}_1$ the bands cross in a certain region of I. Because of the residual interaction, such a crossing does not take place (the no-crossing rule; see Sec. 2.8.4), and we can get a region in which the second derivative is negative:

$$0 > \frac{d^2E}{dJ^2} = \frac{d\omega}{dJ} . \tag{3.9}$$

This means we have an increasing J with decreasing ω, while at the same time the properties of the bands are exchanged. In fact, it is easy to fit such backbending curves by a band mixing calculation with few parameters, one of which is the interaction V between the bands. For small values of V, one obtains a sudden transition which produces backbending, whereas for large values of V the transition region is very broad and no backbending occurs [MR 72, GG 74b].

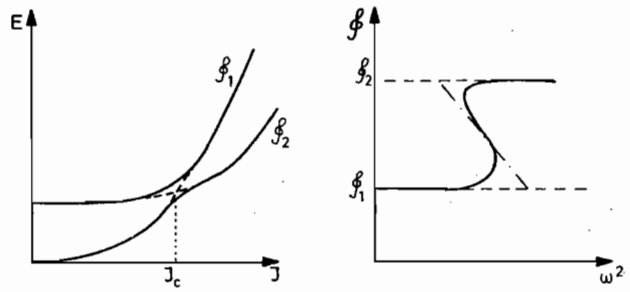


Figure 3.6. Schematic picture of two intersecting bands with different moments of inerita $\mathcal{I}_1$ and $\mathcal{I}_2$, and the corresponding backbending plot. (From [LR 78].)

On the other hand, it is also clear that the strange backbending behavior in the plot of the moment of inertia $\mathcal{I}$ against ω^2 has its origin in the fact that we follow the yrast line in the critical region, that is, we switch over to the crossing band with a different internal structure. If we would stay in the ground state band, which is no longer the yrast band for the large I-values, we would obtain a very smooth behavior for the ω-dependence of the moment of inertia (dashed-line in Fig. 3.6). The reason that one usually follows the yrast band is that these levels are experimentally the most easily accessible.

In all of these considerations, one should, however, keep in mind that these kinds of phenomenological descriptions only give a classification of the spectra and do not say anything about their physical origin. In the case of backbending there is, for instance, the question concerning the nature of the second band. Three types of theoretical interpretations have been given:

(i) In the second band the nucleus has a different deformation, for

instance, a triaxial one. This means that backbending is caused by a sudden change of deformation [Th 73, SV 73].

(ii) The second band is not superfluid, as in the ground state band. This would interpret backbending as a phase transition from a superfluid to a normal-fluid state (Mottelson–Valatin effect [MV 60]).

(iii) The second band is a two quasi-particle band of particles which are rotationally aligned along the axis of rotation (see Sec. 3.3). Then backbending would correspond to a sudden alignment of a pair of nucleons (as proposed by Stephens and Simon [SS 72a]).

The general result of theoretical investigations, which include all these degrees of freedom (see Sec. 7.7), is that for the well deformed nuclei (the classical rotors) the reduction of the pairing correlations is only responsible for the slow change of the moment of inertia at low spin values, but that sudden effects such as backbending are due to alignment of a single high j pair of nucleons. There is also a large number of experimental indications [GSD 73, RSS 74, WBB 75, NLM 76, St 76] for this interpretation of the backbending phenomenon.

In the rare earth nuclei, the $i_{13/2}$ neutrons play an essential role. There is, however, experimental evidence for a second irregularity at spins of $\sim$ 26–30 h [LAD 77, BBB 79a] a so-called *second backbending*, which has been interpreted as the alignment of an $h_{11/2}$ proton pair [FP 78]. Other high j-orbitals may play similar roles in different regions of the periodic table.

3.3 The Particle-plus-Rotor Model

To describe the interplay between the motion of particles and the collective rotation, Bohr and Mottelson [BM 53] proposed to take into account only a few so-called valence particles, which move more or less independently in the deformed well of the core, and to couple them to a collective rotor which stands for the rest of the particles. The division into core and valence particles is not always unique. It is, however, reasonable to use the unpaired nucleon in an odd mass nucleus as a valence nucleon on an even–even core. One also can attribute the particle and the hole of a particle–hole excitation to the valence particles. More generally, one divides the Hamiltonian into two parts: an intrinsic part H_{intr}, which describes microscopically a valence particle or a whole subgroup of particles near the Fermi level; and a phenomenological part H_{coll} which describes the inert core:

$$H = H_{\text{intr}} + H_{\text{coll}}. \tag{3.10}$$

The *intrinsic part* has the form

$$H_{\text{intr}} = \sum_k \epsilon_k a_k^+ a_k + \frac{1}{4} \sum_{klmn} \bar{v}_{klmn} a_k^+ a_l^+ a_n a_m, \tag{3.11}$$

where ϵ_k are single-particle energies in the deformed potential (e.g., Nilsson energies) and $\bar{v}$ is the interaction between the valence particles which is, or can be, neglected in many cases.*

The *collective part* describes the rotations of the inert core:

$$H_{\text{coll}} = \frac{R_1^2}{2\vartheta_1} + \frac{R_2^2}{2\vartheta_2} + \frac{R_3^2}{2\vartheta_3}, \tag{3.12}$$

where the R_i are the body-fixed components of the collective angular momentum of the core. Together with the angular momentum of the valence particles **j** (which is the sum over the single-particle angular momenta) it forms the total angular momentum **I** (see Appendix A):

$$\mathbf{I} = \mathbf{R} + \mathbf{j}. \tag{3.13}$$

Eliminating **R**, H_{coll} can be decomposed into three parts:

$$H_{\text{coll}} = H_{\text{rot}} + H_{\text{rec}} + H_{\text{cor}}, \tag{3.14}$$

where

$$H_{\text{rot}} = \frac{I_1^2}{2\vartheta_1} + \frac{I_2^2}{2\vartheta_2} + \frac{I_3^2}{2\vartheta_3} \tag{3.15}$$

is the pure rotational operator of the rotor [Eq. (1.55)], which acts only on the degrees of freedom of the rotor, that is, the Euler angles. The term

$$H_{\text{rec}} = \sum_{i=1}^{3} \frac{j_i^2}{2\vartheta_i} \tag{3.16}$$

is usually called the recoil term. It represents a recoil energy of the rotor. This operator acts in the coordinates of the valence particles only. For more than one valence particle, it contains a two-body interaction.

Finally, the coriolis interaction

$$H_{\text{cor}} = -\sum_{i=1}^{3} \frac{I_i j_i}{\vartheta_i} \tag{3.17}$$

couples the degrees of freedom of the valence particles to the degrees of freedom of the rotor. This purely kinematic coupling is the only coupling in the model.

It should be noticed that the total angular momentum operators in the laboratory system I_x, I_y, I_z commute with the Hamiltonian (3.10). Although the rotational symmetry is violated in the intrinsic frame (e.g., in the Nilsson Hamiltonian), the model conserves angular momentum for the total system, because the operators I_x, I_y and I_z act only on the Euler angles and commute with the intrinsic components I_1, I_2 and I_3 (see Appendix A). However, it must be emphasized that this rotational invariance is achieved only through the introduction of a phenomenological

*We will see in Chapter 6 that pairing correlations play an important role in deformed nuclei and we should therefore talk about quasi-particles rather than particles. This only yields simple occupation factors to the formulae shown below, which we give sometimes for the sake of completeness, however. For a deeper understanding, the reader is referred to Chapter 6.

core. If one wants to treat all particles microscopically (the limit of vanishing core), the Euler angles are redundant variables (see Sec. 11.3).

The wave function of the system may be written as

$$|\Psi^I_M\rangle = \sum_K \Phi_K |IMK\rangle, \tag{3.18}$$

where Φ_K depends on the coordinates of the valence particles and the $|IMK\rangle$ depend on the Euler angles and are defined in Eq. (A.21). In the simplest case, there is only one valence particle and an axially symmetric rotor. Using this model a large number of the experimental spectra of odd nuclei has been reproduced very accurately, and from this point of view it is certainly one of the most powerful models in nuclear physics. However, until now a clear-cut microscopic derivation has been missing. Such a derivation should start from a many-body Hamiltonian (2.19). The set of $3A$ particle coordinates should be transformed in a proper way into a set of $3A-3$ internal coordinates and 3 Euler angles which describe the collective motion. To this transformation should correspond a decomposition of the Hamiltonian into an internal and a collective part, as in Eq. (3.10). Finally, one could hope to describe the internal motion by a deformed shell model.

Many attempts have been made in this direction and we give a short discussion of them in Section 11.3, but up to the present time the problem has still not been completely solved. In the case of well deformed nuclei the model can be backed by the following arguments.

(i) Microscopic Hartree–Fock calculations show very pronounced minima in the energy surface at axial symmetric deformations, which justifies the notion of a rotating core.

(ii) The model is to some extent equivalent to the cranking model (see Sec. 3.4), which is microscopically founded at least in the limit of strong nearly axial symmetric deformations.

(iii) Villars and Cooper [VC 70] have shown that, in introducing redundant coordinates, a Hamiltonian of a form similar to that in Eq. (3.10) can be found with, however, additional coupling terms. In the limit of strong deformation and with further assumptions, they obtained the correct expression for the moment of inertia.

The particle-plus-rotor model has, however, also been applied with great success to nuclei in the region of small deformations and to transitional nuclei. It is not clear at the present time how this can be explained microscopically.

3.3.1 The Case of Axial Symmetry

Assuming that the rotor has the 3-axis as axis of symmetry, that is, $\vartheta_1 = \vartheta_2 = \vartheta$, there can be no collective rotation around this axis and the 3-component of $\mathbf{R}$ has to vanish (see Sec. 1.5). From Eq. (3.13) it follows

immediately that K, the 3-component of the total angular momentum **I**, has to be equal to Ω, the 3-component of **j**:

$$K=\Omega. \tag{3.19}$$

For the different terms of the Hamiltonian (3.10), (3.14), we obtain in this case

$$H_{\text{intr}}=\sum_{i,\Omega}\epsilon^i_\Omega a^+_{i\Omega}a_{i\Omega}, \tag{3.20}$$

$$H_{\text{rot}}=\frac{\mathbf{I}^2-I_3^2}{2\mathcal{I}}, \tag{3.21}$$

$$H_{\text{rec}}=\frac{1}{2\mathcal{I}}(j_1^2+j_2^2), \tag{3.22}$$

$$H_{\text{cor}}=-\frac{1}{\mathcal{I}}(I_1j_1+I_2j_2)=-\frac{1}{2\mathcal{I}}(I_+j_-+I_-j_+). \tag{3.23}$$

In (3.20) we have neglected the residual interaction. The single-particle levels in the axially symmetric well are labeled by $k=(i,\Omega)$ and the corresponding eigenfunctions will be denoted by Φ^i_Ω.

The recoil term acts only in the intrinsic coordinates. It is often neglected, because the intrinsic single-particle energies ϵ^i_Ω are adjusted to experimental data. We follow this argument in the following discussion and omit H_{rec} for the moment. Only in Section 3.3.2.1 will we take it up again. The Hamiltonian (3.20–3.23) has eigenfunctions of the type (3.18), which can be found by a numerical diagonalization of the Hamiltonian (3.10). However, the different terms in (3.20–3.23) are of different importance, depending on the physical situation. Therefore, it is useful to consider three limits in which one of the terms becomes predominant and which as a consequence can be solved analytically (for a review, see [St 75a]):

In the *strong coupling limit*, the odd particle adiabatically follows the rotations of the even mass core. It is realized if the coupling to the deformation is much stronger than the perturbation of the single particle motion by the Coriolis interaction.

In the *weak coupling limit*, which is realized for very small deformations, the odd particle essentially moves on spherical shell model levels only slightly disturbed by, for example, the quadrupole vibrations (see Sec. 9.3.3).

In the *decoupling limit*, the Coriolis force is so strong that the coupling to the deformation of the core may be neglected. The total angular momentum and the single-particle angular momentum are then parallel to one another.

3.3.1.1 The Strong Coupling Limit (Deformation Alignment). The strong coupling limit is realized when the Coriolis interaction matrix elements are small compared with the level splitting of the single-particle energies in the deformed shell model for different values of Ω. We should expect that this

is the case

(i) for large deformations β, because the level splitting in the Nilsson Hamiltonian is $\propto \beta$, whereas the rotational constant $\hbar^2/2\mathcal{I}$ is, according to a simple empirical rule (Eq. (1.50), $\propto \beta^{-2}$; and

(ii) for small Coriolis matrix-elements. As we shall see in Eq. (3.33), they are $\propto [(I(I+1)-K^2)(j(j+1)-\Omega^2)]^{1/2}$, that is, they are small either for low spins I or for nucleons in orbitals with small particle angular momenta j. For large j-values they can only be neglected for high Ω values.

It is called the *strong coupling* or *deformation aligned* limit because in this case K is a good quantum number. The angular momentum **j** of the valence particles is strongly coupled to the motion of the core. In a semiclassical picture, **j** precesses around the 3-axis, which is shown in the coupling scheme of Fig. 3.7a. Since H_{cor} is the only term in the Hamiltonian which couples the particle and rotor degrees of freedom, the eigenfunctions are in this limit products of the functions Φ^i_K (eigenfunctions of H_{intr}, e.g., the single-particle Nilsson functions) and the eigenfunctions of the symmetric rotor $|IMK\rangle$ (see Appendix A).

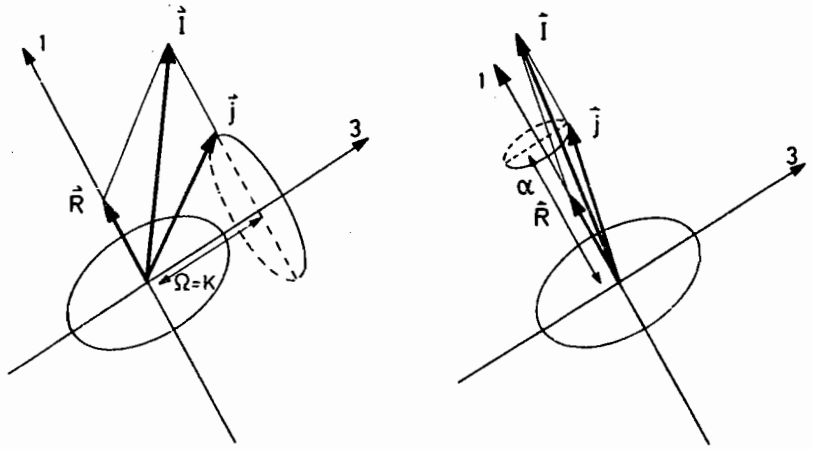


Figure 3.7. Coupling schemes in the particule-plus-rotor model: (a) strong coupling; (b) rotational alignment.

We have seen in Section 1.5.1 that the Hamiltonian (3.10) has an additional symmetry, $\mathcal{R}_1$, which describes a rotation of the core by 180° around the 1-axis of the body-fixed system

$$\mathcal{R}_1 = e^{i\pi R_1} = e^{i\pi I_1} e^{-i\pi j_1}. \tag{3.24}$$

We therefore have to symmetrize the wave function (3.18) and get together with Eq. (A.24) the following set of eigenfunctions [Ke 56]

$$|\Psi^I_{MK}\rangle \frac{1}{\sqrt{2(1+\delta_{K0})}} \{\Phi^i_K |IMK\rangle + (-)^I e^{-i\pi j_1} \Phi^i_K |IM-K\rangle\}. \tag{3.25}$$

If we know the decomposition of Φ^i_K into eigenstates of $\mathbf{j}^2$

$$|\Phi^i_{K=\Omega}\rangle = \sum_{nj} C^i_{nj}|nj\Omega\rangle, \tag{3.26}$$

we find

$$e^{-i\pi j_1}|\Phi^i_{K=\Omega}\rangle = \sum_{nj} C^i_{nj}(-)^{-j}|nj-\Omega\rangle. \tag{3.27}$$

The energy spectrum which corresponds to (3.25) is given by

$$E^i_K(I) = \epsilon^i_K + \frac{1}{2\mathcal{I}}\left(I\cdot(I+1)-K^2\right) \tag{3.28}$$

[usually, instead of ϵ^i_K in (3.28), we have quasi-particle energies E^i_K as they are defined in Eq. (6.72)]. This is the spectrum of a rotational band. The lowest possible spin is $I_0 = K$. The bandhead $E^i_K(I_0)$ is not precisely ϵ^i_K, but shifted a little, especially if we take into account the recoil term and the residual interaction. The spectrum has a spacing $\Delta I = 1$ and its moment of inertia is that of the rotor.

In this strong coupling limit we have neglected the Coriolis interaction completely. Taking it into account at least in first order perturbation theory, we get a contribution only for $K = 1/2$ bands

$$E^i_{K=1/2}(I) = \epsilon^i_{K=1/2} + \frac{1}{2\mathcal{I}}\left\{I(I+1) - \frac{1}{4} + a^i\left(I+\frac{1}{2}\right)(-)^{I+1/2}\right\}, \tag{3.29}$$

where the so-called *decoupling factor* is given by

$$a^i = i\langle\Phi^i_{1/2}|j_+ e^{-i\pi j_1}|\Phi^i_{1/2}\rangle \tag{3.30}$$

or if Φ^i_K is of the form (3.26),

$$a^i = -\sum_{nj}|C^i_{nj}|^2(-)^{j+1/2}(j+\tfrac{1}{2}). \tag{3.31}$$

This means, for example, that for a positive decoupling factor (major components with $j+\frac{1}{2}$ odd) the levels with odd values of $I+\frac{1}{2}$ ($I=\frac{1}{2}, \frac{5}{2}, \frac{7}{2}, \cdots$) are shifted downwards. This explains very nicely the rather distorted bands for $K=\frac{1}{2}$ in many nuclei [ABH 56] where there are, in fact, two bands having $\Delta I = 2$ each (even and odd values of $I+\frac{1}{2}$) shifted against one another. The reason for this decoupling comes from the symmetrization Eq. (3.25) with respect to $\mathcal{R}_1$ mixing states with $K=\frac{1}{2}$ and $K=-\frac{1}{2}$ via the Coriolis matrix elements. The strength of the decoupling factor a^i depends on the j-components which contribute to the single-particle wave function $\Phi^i_{1/2}$. If levels with large single-particle angular momenta are involved, it is very strong. However, perturbation theory is no longer valid in such cases, and also other matrix elements of the Coriolis operator come into play. Therefore, the strong coupling limit is no longer realized and we have to consider a different limit.

3.3.1.2 The Weak Coupling Limit (No Alignment). As we have said, the strong coupling approximation breaks down if the Coriolis matrix elements are no longer negligible compared to the energy splitting of the single-

particle levels belonging to different K values. Let us therefore study the Coriolis matrix elements in more detail.

$$\langle \Psi^I_{MK+1}|H_{\text{cor}}|\Psi^I_{MK}\rangle = -\frac{1}{\mathcal{I}}\sqrt{I(I+1)-K(K+1)}\ \langle \Phi_{\Omega+1}|j_x|\Phi_\Omega\rangle. \quad (3.32)$$

If Φ_Ω is of the form shown in Eq. (3.26), we obtain

$$\langle \Psi^I_{MK+1}|H_{\text{cor}}|\Psi^I_{MK}\rangle = -\frac{1}{2\mathcal{I}}\sum_{nj}|C_{nj}|^2\sqrt{I(I+1)-K(K+1)}\ \sqrt{j(j+1)-\Omega(\Omega+1)}\ . \quad (3.33)$$

We see that these matrix elements are large for large values of I/K and j/Ω. That is, if, for example, levels with large j and small Ω values are involved. Particles in such levels have high angular momentum and a density distribution close to the 3-axis. Therefore, it is clear that a rotation of the core perpendicular to this axis has a great influence on the motion of these particles.

A well known example is the neutron $1i_{13/2}$ level, which lies in the vincinity of the Fermi level for light rare earth nuclei such as Dy and Er. One can estimate the Coriolis matrix element for the $\Omega=\frac{1}{2}$ case to be $0.1\times I$ [MeV], which is in fact quite large compared with the level spacing of H_{intr}. Since such levels with high j-values are drastically shifted downwards by the spin orbit term of the shell model (see Sec. 2.4) into a shell with a different N-quantum number, these levels are rather pure configurations, that is, $C_{nj}\simeq 1$ (intruder state). It is therefore sufficient in the following to consider *only one such single j-shell*. The $1i_{13/2}$ shell is not the only such case. The energy of the largest j-value in each major shell is lowered drastically and has an important role in many rotational spectra.

Vogel [Vo 70] proposed a limit (usually called weak coupling or no alignment limit) in which the K-splitting of H_{intr} is totally neglected. (This, of course, can only be a valid approximation for small deformation.) Now $\mathbf{j}^2$ and $\mathbf{R}^2$ commute with H_{intr} and we can construct eigenfunctions of $\mathbf{R}^2$ and R_3 (the eigenvalue of the latter is, of course, zero). A proper angular momentum coupling gives*

$$|\Psi^{IR}_M\rangle = \sum_K (-)^{j-K} C^I_{K\ -K\ 0}{}^{j\ R}\Phi_K|IMK\rangle. \quad (3.34)$$

These wave functions diagonalize $\mathbf{R}^2$, and the corresponding spectrum is of the form

$$E(I) = E_{\text{intr}} + \frac{1}{2\mathcal{I}}R(R+1), \quad (3.35)$$

with

$$|j-R| \leqslant I \leqslant j+R;\ R=0,2,4,\cdots (\mathfrak{R}_1 \text{ symmetry}). \quad (3.36)$$

This means that for each rotational quantum number R, $\mathbf{j}$ can have $2j+1$ orientations without changing the energy of the system (zero cou-

* Since we couple the angular momenta $\mathbf{I}$ and $-\mathbf{j}$ to $\mathbf{R}=\mathbf{I}-\mathbf{j}$, we have to apply a coupling rule as in the *ph*-case of Eq. (2.46).

pling). The splitting of the $2j+1$ levels can be taken into account in first-order perturbation theory (we will take a Nilsson term $\sim\beta r^2 Y_{20}$ as perturbation).

$$E(I)=E_0+\frac{1}{2\mathcal{J}}R(R+1)-\beta\hbar\omega_0\langle\Psi_M^{IR}|r^2Y_{20}|\Psi_M^{IR}\rangle. \tag{3.37}$$

To each orientation of **j** there is a whole rotational band of the core with $\Delta R=2$. The levels with the highest I values $I=R+j$ for a given energy (i.e., for a given R) correspond to the yrast levels. These levels are connected by strong $E2$ transitions [Re 75a] and are experimentally the most easily accessible. They are called *favored states** [St 75a] and their energy is given by

$$E(I)=E_{\text{intr}}+\frac{1}{2\mathcal{J}}(I-j)(I-j+1). \tag{3.38}$$

This means that these states lie on a parabola with a minimum at $I\simeq j$, which is experimentally widely confirmed, an example of which is shown in Fig. 3.8.

* Levels with a lesser degree of alignment (e.g. $I=j+R-1$) are called *unfavored states*.

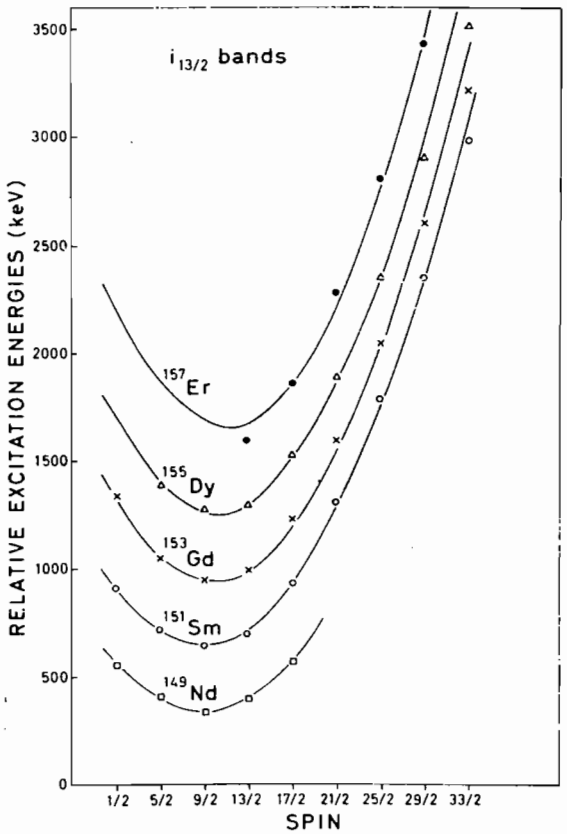


Figure 3.8. Relative excitation energies as a function of spin for the $i_{13/2}$ bands in some $N=89$ nuclei. (The experimentally observed band members are plotted such that the $13/2^+$ states are equally spaced (300 keV).) (From [LR 76].)

On the right branch of this parabola **j** and **R** are aligned (**j** ∥ **R**),

$$I=j+R=j, j+2, j+4, \ldots,$$

and on the left branch **j** and **R** are anti-aligned (**j** ∥ −**R**); the core rotates in the opposite direction as **j***:

$$I=j-R=j-2, j-4, \ldots, \tfrac{3}{2} \text{ (or } \tfrac{1}{2}).$$

As an empirical rule one can say that the weak coupling limit is strictly valid only for very small deformations $|\beta A^{2/3}|<4$. We shall see in the next section, however, that the states with **I** ∥ **j** are energetically favoured even in the case of stronger deformations, where neglecting the Ω-dependence of H_{intr} is no longer justified. Therefore, the formula (3.38) is also valid for many levels in more deformed nuclei (see for instance Fig. 3.8) in the so-called rotation aligned coupling scheme.

3.3.1.3 The Decoupling Limit (Rotational Alignment). In the case of intermediate deformations, the energy splitting in the intrinsic part of the Hamiltonian can no longer be neglected. In this case, the orientation of the external large j-particle is no longer independent of the motion of the core. Stephens et al. [SDN 73] realized that the requirement of maximal overlap of the single-particle density distribution (which is concentrated mostly in a plane perpendicular to j) with the core can be fulfilled if the external particle is aligned along the rotational axis of a prolate nucleus (see also Sec. 2.8.4).

Mathematically, we can understand this in the model of a single j-shell ($C_{nj}\simeq 1$, like, for example, in the $i_{13/2}$ case). Neglecting $\mathbf{l}^2$ and $\mathbf{ls}$-term, the Nilsson energies are simply given by the diagonal matrix element

$$\begin{aligned}\epsilon_\Omega &= \epsilon_0-\beta\cdot\hbar\omega_0\langle nlj\Omega|r^2Y_{20}|nlj\Omega\rangle \\ &= \epsilon_0+\beta k\frac{3\Omega^2-j(j+1)}{4j(j+1)}=\epsilon_0-\frac{1}{4}\beta k+C\Omega^2,\end{aligned} \tag{3.39}$$

where k and C do not depend on Ω. Also, the recoil term (3.22) can be calculated as a diagonal matrix element and yields

$$H_{\text{rec}}=\frac{1}{2\mathcal{I}}\{j(j+1)-\Omega^2\}. \tag{3.40}$$

Therefore, the Hamiltonian (3.10) in this approximation is:

$$H=\epsilon_0-\frac{1}{4}\beta k+\frac{1}{2\mathcal{I}}(I(I+1)+j(j+1))+\left(C-\frac{1}{\mathcal{I}}\right)\hat{I}_3^2+H_{\text{cor}}. \tag{3.41}$$

In a certain region of deformation, where $C\simeq 1/\mathcal{I}$, there is thus a cancellation of the K dependence coming from the intrinsic and rotational parts of the Hamiltonian. This turns out to apply for a rather broad domain of intermediate deformations. Unfortunately, the eigenfunctions of H_{cor} which we needed to solve (3.41) cannot be given analytically in the general case. Thus, we wish to give a qualitative discussion. Writing the

* In principle, there are also anti-aligned states with $R>j$ [PO 78].

Coriolis term in the form*

$$H_{\text{cor}} = -\frac{1}{\mathcal{I}} \mathbf{I}_{\perp} \mathbf{j}_{\perp}, \tag{3.42}$$

we see that it has its lowest expectation value for the wave functions with $\langle \mathbf{I} \rangle \parallel \langle \mathbf{j} \rangle$. Since $\mathbf{R}$ is perpendicular to the 3-axis, one gets the lowest eigenvalue for wave functions with a single-particle angular momentum aligned along a rotational axis perpendicular to the symmetry axis. For this purpose, we choose the 1-axis and accordingly construct eigenfunctions $|\tilde{\Phi}_{\alpha}\rangle$ of j_1 with eigenvalues α by rotating the eigenfunctions $|\Phi_K^j\rangle$ of j_3 through 90° about the 2-axis:

$$|\tilde{\Phi}_{\alpha}\rangle = \sum_K d^j_{K\alpha}\left(-\frac{\pi}{2}\right)|\Phi_K^j\rangle, \tag{3.43}$$

where $d^j_{K\alpha}$ is the Wigner function of this rotation [Ed 57].

For the total wave function it is, however, not so easy to make such a construction in which α is a good quantum number. The ansatz

$$|\Psi^I_{M\,\alpha}{}^{j}\rangle = \sum_K d^j_{K\alpha}\left(-\frac{\pi}{2}\right)\Phi_K^j|IMK\rangle \tag{3.44}$$

is an eigenfunction of H_{cor} at least for $I \gg K$. With

$$I_-|IMK\rangle = \sqrt{I(I+1)-K(K+1)}\;|IMK+1\rangle \simeq I|IMK+1\rangle \tag{3.45}$$

we then easily get

$$H_{\text{cor}}|\Psi^I_{M\,\alpha}{}^{j}\rangle = -\frac{1}{\mathcal{I}} I\cdot\alpha|\Psi^I_{M\,\alpha}{}^{j}\rangle \qquad (I \gg K), \tag{3.46}$$

which shows that α is the projection of $\mathbf{j}$ onto $\mathbf{I}$.

From Eq. (A24) and [Ed 57, Eq. (4.2.4)] we obtain

$$\mathcal{R}_1|\Psi^I_{M\,\alpha}{}^{j}\rangle = (-)^{I-\alpha}|\Psi^I_{M\,\alpha}{}^{j}\rangle, \tag{3.47}$$

which means that $I-\alpha$ has to be even in order to fulfill the symmetry condition $\mathcal{R}_1 = 1$.

In the case of $C \simeq 1/\mathcal{I}$, we find for the spectrum of the Hamiltonian (3.41):

$$E(I,\alpha) = \text{const.} + \frac{1}{2\mathcal{I}}(I(I+1)+j(j+1)-2I\alpha)$$

$$= \text{const.} + \frac{1}{2\mathcal{I}}(I-\alpha)(I-\alpha+1) \tag{3.48}$$

$$= \text{const.} + \frac{1}{2\mathcal{I}}R(R+1) \tag{3.49}$$

where $R = I-\alpha = 0, 2, 4, \ldots$ has to be even because of the symmetry condition (3.47).

The lowest lying states are therefore the ones which are maximally aligned ($\alpha = j$, *favored states*). This corresponds exactly to the picture of a valence particle whose spin is oriented perpendicular to the 3-axis while

*$\mathbf{I}_{\perp}$ and $\mathbf{j}_{\perp}$ are the components of $\mathbf{I}$ and $\mathbf{j}$ perpendicular to the symmetry axis.

the core is completely decoupled and rotates with $R = I - \alpha$, giving rise to a spectrum $\Delta I = 2$, but which is otherwise equivalent to that of the neighboring even nucleus. Such bands have been seen, for example, in weakly deformed nuclei (see Figs. 3.8 and 3.9).

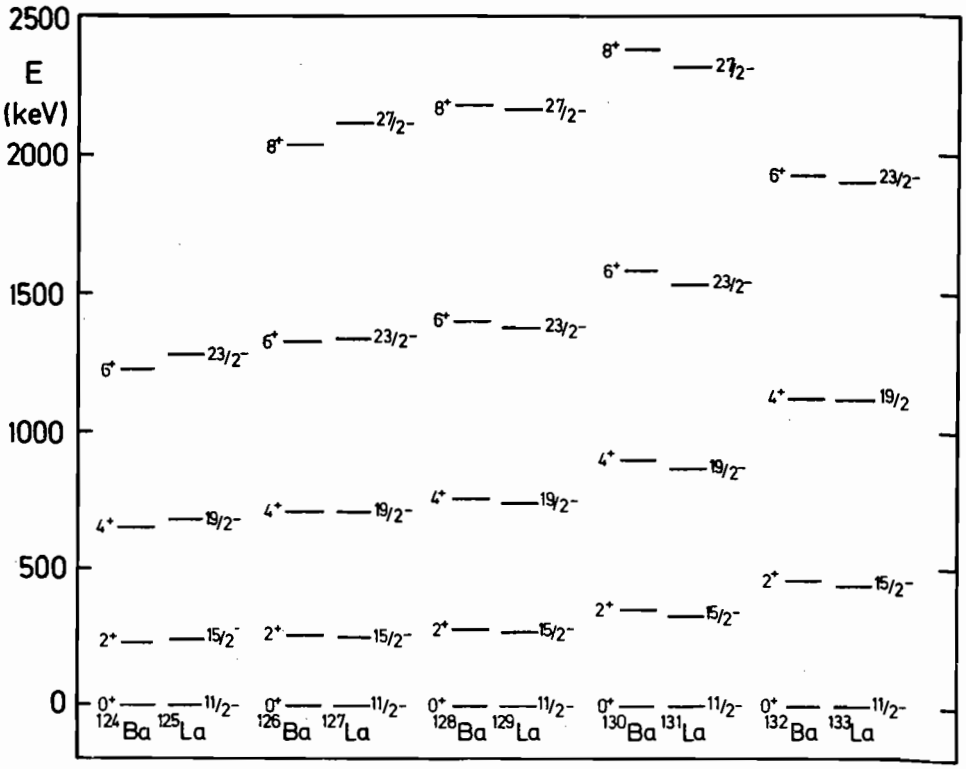


Figure 3.9. Comparison of ground band levels in some Ba isotopes with negative parity bands in the neighboring La nuclei. In most cases the La 11/2 level is not the ground state, and its energy has been subtracted from all levels shown for that isotope. (From [SDL 72].)

Lesser aligned states (*unfavored states*) have $\alpha = j - 1$. They often lie at higher energies and are then not populated in (HI, xn) reactions.

Figure 3.10 shows the exact solution of the axially symmetric particle-plus-rotor model for one valence nucleon in a $1h_{11/2}$ shell as a function of the deformation β. For very small β-values, one has the weak coupling scheme of several nearly degenerate multiplets. On the oblate side ($\beta < 0$) the strong coupling is realized. In this case, the lowest level in the Nilsson scheme (see Fig. 2.21.) is the $\Omega = 11/2$ level. Its Coriolis matrix element is very small and a strongly coupled band with $\Delta I = 1$ is observed. On the prolate side the opposite is true: The lowest level has $\Omega = \frac{1}{2}$ and a very large Coriolis matrix element. The yrast band (the levels with the highest I-values) is now formed by a decoupled band $I = 11/2, 15/2, 19/2, \ldots$ with $\Delta I = 2$ and a level spacing, which is more or less that of the rotor (as seen at $\beta = 0$). We see also that for these completely aligned yrast levels the rotation aligned coupling scheme is very well realized over a wide range of

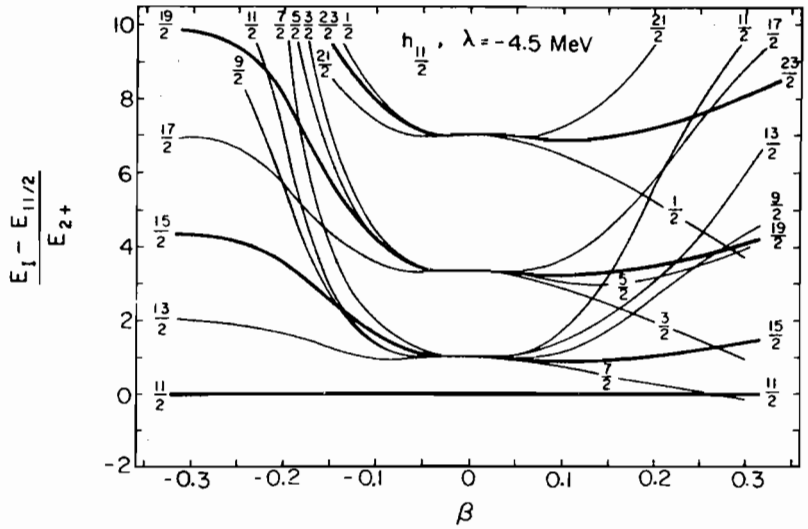


Figure 3.10. The eigenvalues of an axially symmetric particle-plus-rotor model, with one particle sitting in the $1h_{11/2}$ shell, as a function of the deformation β. Plotted are the excitation energies over the $I = 11/2$ level. The Fermi surface λ is below the entire $h_{11/2}$ orbital. (From [St 75a].)

intermediate β-values: $0.13 \sim \beta < 0.23$. Only at very large deformations does one find deviations.

The structure of the bands also depends, of course, on the position of the Fermi level. We had here the simplest case of only one particle in the shell. The complete analogue is one hole in the high j-shell. There the situation is reversed: On the prolate side are the levels with large Ω-values —that is, one observes a strongly coupled scheme—and on the oblate side are the levels with small Ω-values and a decoupled scheme.

These considerations show that the structure of the rotational bands built on such high j levels provides an excellent tool to distinguish experimentally between prolate and oblate deformations in the transition region.

Summarizing the results of this section, we can say that

(i) In many cases, the Coriolis interaction can be neglected. Then the valence nucleons rotate around the symmetry axis of the core and change their orientation with it (strong coupling $\Delta I = 1$).

(ii) In cases of small deformation and strong Coriolis interaction, the valence nucleons orient their angular momentum more or less independently of the orientation of the core. The core rotates with $\Delta I = 2$ (weak coupling).

(iii) For intermediate deformations it can happen by a cancellation effect that a rotational alignment takes place. The valence particles orient their angular momentum parallel to the collective angular momentum (perpendicular to the symmetry axis of the core) and again we have $\Delta I = 2$.

3.3.2 Some Applications of the Particle-plus-Rotor Model

The model of a few valence particles coupled to a symmetric rotor has been used in a very large number of cases to fit the experimental rotational bands. In general, it has been very successful. We do not want here to go into the many different versions which have been used but wish to discuss a few characteristic examples.

3.3.2.1 Strongly Mixed Bands in Well-Deformed Odd Mass Nuclei. The spectra of odd mass nuclei in the deformed region show many rotational bands which reveal very nicely the strong coupling picture. Only in cases where the particle sits in a single-particle shell with a high j-value (e.g., the $1i_{13/2}$ shell in the Er region), one also observes very distorted bands.* For small I values they start out like strongly coupled bands with $\Delta I = 1$, but soon we observe staggering: The levels with $I + 1/2$ even are shifted more and more against the levels with $I + 1/2$ odd, so that in the end we have two separate bands with $\Delta I = 2$ (see the positive parity band in Fig. 3.11 and [BDL 75]).

* For a comparison of these distorted bands with the backbending behavior in neighboring even nuclei, see [SKS 74].

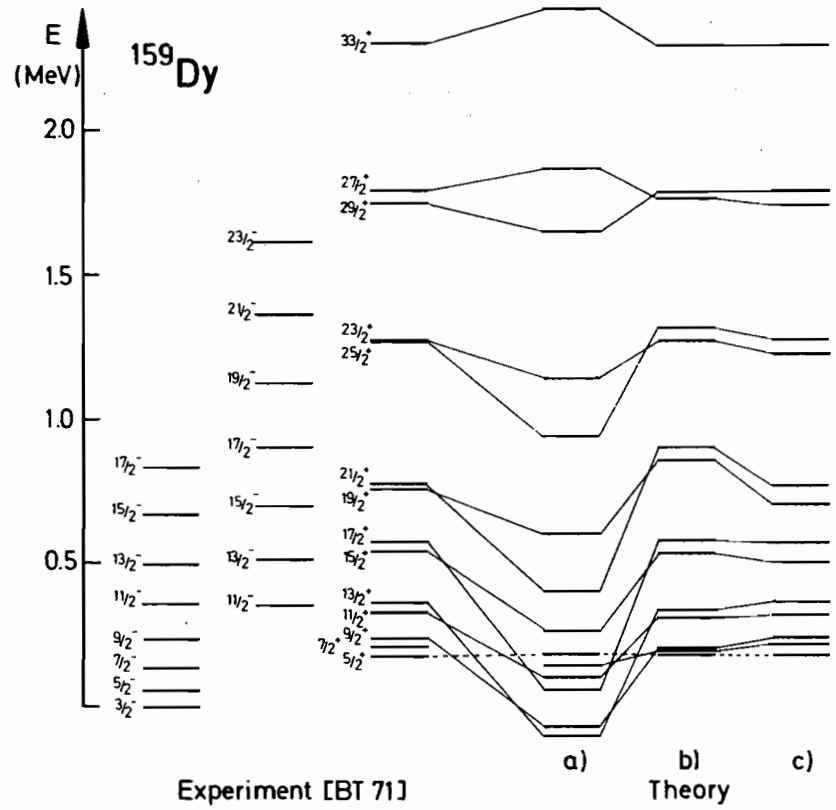


Figure 3.11. Three rotational bands in the nucleus ^{159}Dy. The $3/2^-$ and the $11/2^-$ bands are strongly coupled ($\Delta I = 1$). The positive parity band with the band head $I = 5/2$ is strongly perturbed, as discussed in the text. It is compared with three calculations: (a) a particle-plus-rotor fit without attenuation ρ, (b) a particle-plus-rotor fit with attenuation $\rho = 0.5$, and (c) a self-consistently determined attenuation according to Eq. (3.56). (From [Ri 77].)

To describe these bands one usually diagonalizes the Hamiltonian (3.10) in the strong coupling basis (3.25). The single (quasi-) particle energies are taken from a Nilsson model and the parameters like the deformation β (and the gap Δ, see Chap. 6) are used to fit the spectrum. It thereby turns out that one can only reproduce the experimental spectrum if one introduces an additional parameter ρ—the *attenuation factor*—which weakens the Coriolis interactions, that is, if we use $\rho \cdot H_{\text{cor}}$ instead of H_{cor} [HRH 70, LRB 72, HK 77]. The original Coriolis force turns out to be much too strong and it has to be reduced by a factor $\rho \sim 0.4-0.8$. With this ad hoc attenuation and the other parameters of the model reasonably chosen, we are able to reproduce the distorted bands and find that for low I-values, that is, for small Coriolis interaction, one is in the strong coupling limit with $\Delta I = 1$. For higher I-values, however, the Coriolis force gets stronger and aligns the odd particle with the large j-value parallel to the rotational axis. For I-values $I \gtrsim j$, we have a rotational aligned motion, that is, large K-mixing and a splitting of the band in favored states with maximal alignment $\alpha = j$ for $(I-j)$ even, and unfavored states with lesser alignment $\alpha = j-1$ for $(I-j)$ odd. For the same value of angular momentum R of the core, which is given by $R = I - \alpha$, these two bands are almost degenerate (see Fig. 3.11).

Several attempts have been made to explain the *attenuation factors* as a kind of effective charge in the linear response approach [BPC 72, HK 75]. It turns out that this problem is closely connected with a proper treatment of the recoil term (3.22). The argument, that it is already taken into account in the fit of the band head energies, does not apply because in the one-particle case it is proportional to $j^2 - K^2$ and therefore has a strong K-dependence which shifts the band heads [ORG 75].

To understand qualitatively the effect of the recoil term we restrict ourselves to the case of only one particle and rewrite the particle-plus-rotor Hamiltonian (3.20ff) in the following way [Kr 79].

$$H = H_{\text{intr}} + \frac{1}{2\mathcal{I}}\left(I(I+1) - \mathbf{j}^2\right) + H'_{\text{cor}} \tag{3.50}$$

with the new Coriolis term

$$H'_{\text{cor}} = -\frac{1}{\mathcal{I}}(\mathbf{I}-\mathbf{j})\mathbf{j}_{\perp} = -\frac{1}{\mathcal{I}}\mathbf{R}\mathbf{j} = -\omega\mathbf{j}, \tag{3.51}$$

where the collective angular velocity is given by

$$\omega = -\mathbf{R}/\mathcal{I}. \tag{3.52}$$

The Coriolis term H'_{cor} in Eq. (3.51) is attenuated compared to H_{cor} (3.23). To see this we go into the limit, where the odd particle is nearly aligned to the collective rotation **R**. We then have

$$\mathbf{R} \simeq \mathbf{I}_{\perp}\left(1 - \frac{\alpha}{I}\right) \tag{3.53}$$

and find that H'_{cor} is in this case proportional to H_{cor}:

$$H'_{\text{cor}} = \left(1 - \frac{\alpha}{I}\right) H_{\text{cor}} = \rho \cdot H_{\text{cor}} \tag{3.54}$$

with an I-dependent attenuation factor

$$\rho = 1 - \frac{\alpha}{I}. \tag{3.55}$$

Only for very high I-values does the attenuation disappear.

So far, the discussion of the influence of the recoil term has been restricted to the pure single-particle part of this operator. In fact, it also contains a two-body

term $\mathbf{j}_{ik} \cdot \mathbf{j}_{lm} a_i^+ a_l^+ a_m a_k$, which produces an interaction with the particles of the core. We can treat it in the mean field approximation [Ri 77] and end up with a self-consistent attenuation factor

$$\rho = 1 - \frac{\langle j_1 \rangle}{\sqrt{I(I+1) - \langle j_3^2 \rangle}} . \tag{3.56}$$

At the limit of alignment it goes over into the form (3.55). We can also show that this approximation is equivalent to the cranking model (see Sec. 3.4). In a microscopic treatment of rotational bands in odd mass nuclei within the self-consistent cranking model (see Sec. 7.7), the attenuation of the Coriolis interaction is automatically incorporated [RMB 74, RM 74].

3.3.2.2 Backbending in Even Nuclei. The backbending phenomenon (see Sec. 3.2.4.) has been explained by Stephens and Simon [SS 72a] as an alignment of two neutrons in the $1i_{13/2}$ shell. If these two neutrons, instead of rotating around the 3-axis, align along the rotation axis of the nucleus, this adds an additional $13/2 + 11/2 = 12$ units of angular momentum. Therefore, the nucleus can decrease its collective rotation while increasing its total angular momentum through the addition of single-particle angular momentum.

To describe this idea mathematically, Stephens and Simon diagonalized the Hamiltonian (3.10) (with attenuation of the Coriolis term) taking as the basis the Slater determinant of the unperturbed core

$$|\Phi_0\rangle = |IMK = 0\rangle$$

and two quasi-particle excitations (see Chap. 7)

$$|\Phi_K\rangle = \beta_{K_1}^+ \beta_{K_2}^+ |IMK = K_1 + K_2\rangle, \tag{3.57}$$

where β_K^+ is a creation operator for a quasi-particle in the state $\Omega = K$ of the $1i_{13/2}$ shell.

For low I values, the yrast states are given mainly in the zero quasi-particle state $|\Phi_0\rangle$ of the pure rotor. The excited bands are two quasi-particle bands. For higher I-values, the particles align their angular momenta along the axis of rotation, one of them to $\alpha = j = 13/2$, the second to $\alpha = j - 1 = 11/2$. Thus, we find a mixing of the states Φ_0 and Φ_K ($K = 0, 1, \ldots$) and arrive, at the limit of full alignment, at a two-quasi-particle state of the form [see Eq. (3.44)]:

$$|\Phi\rangle_{al} = \sum_{K_1 K_2} d^j_{K_1 j}\left(-\frac{\pi}{2}\right) d^j_{K_2 j-1}\left(-\frac{\pi}{2}\right) \beta_{K_1}^+ \beta_{K_2}^+ |IMK_1 + K_2\rangle. \tag{3.58}$$

It can be written schematically as $\tilde{\beta}_j^+ \cdot \tilde{\beta}_{j-1}^+ |\Phi_0\rangle$, where the $\tilde{\beta}_\alpha^+$ are the quasi-particle operators quantized along the 1-axis.

Stephens and Simon [SS 72a] were able to reproduce the experimental backbending spectra reasonably well with this method. Since they did not take into account, however, the gradual change in the pairing correlations caused by the Coriolis–anti-pairing effect, they could not obtain the deviations from the $I \cdot (I+1)$ law at low spin values.

In spite of the fact that this model describes the important effect of two aligning particles properly, it does not allow us to decide whether there is any other mechanism which could be the origin of the observed backbending phenomenon. To decide whether it is caused by a change in shape, by a phase transition to a normal fluid state, or by an alignment process, one has to carry out a microscopic calculation which allows for all these degrees of freedom. Such investigations have been done (see Sec. 7.7). They show that the rotational alignment of two particles is

the most important effect. These processes are found in all nuclei in which a high j-sheil exists in the vicinity of the Fermi surface. However, it is another question as to whether they produce backbending. This depends finally on the strength of the interaction between the two crossing bands: Only for rather small coupling matrix elements do we observe a sudden transition, that is backbending.*

3.3.3 The Triaxial Particle-plus-Rotor Model

We have already seen in Section 1.5.3 that Davydov et. al. [DF 58] used a triaxial rotor to explain the low lying 2^+ states in some transitional nuclei. The model can also be extended to odd mass nuclei by the coupling of an external particle to a triaxial rotor [Pa 61, HS 62, PR 62, PS 65, MSD 74, Me 75, FT 75, TF 75, Le 76, DF 77, LLR 78]. It has been applied to cases where the external particle sits in a high j-shell, and has turned out to be very powerful as a description of energy levels and decay schemes of many transitional nuclei.† However, at present the microscopic foundation is missing. Using microscopic theories, the calculations of static energy surfaces in these mass regions show no pronounced minima at triaxial deformations, which would justify this simple picture. In fact, there exist other models based on a *vibrational picture* [AP 76, PVD 77, YNN 76] that are also able to reproduce such spectra. It is not clear at present whether there is any connection between these two pictures of transitional nuclei.

We restrict ourselves in the following discussion to one external particle in a high j-shell (e.g., $h\ 11/2$) and couple it to a triaxial rotor. In this case, the Hamiltonian has the form [MSD 74]:

$$h = \sum_{i=1}^{3} \frac{R_i^2}{2\vartheta_i} + h_0 + kr^2\beta \left\{ \cos\gamma Y_{20} + \sin\gamma \frac{1}{\sqrt{2}} (Y_{22} + Y_{2-2}) \right\}. \tag{3.59}$$

The β, γ-dependence of the moment of inertia ϑ_i is that for irrotational flow [eq. (1.48)] and only the overall constant is adjusted. The constant k is given by the splitting of the j-shell in the Nilsson scheme. h_0 is the spherical harmonic oscillator. Usually a single-particle pairing field with constant gap Δ is also taken into account.

Figure 3.12 shows the spectrum of the Hamiltonian (3.59) as a function of γ at a typical deformation $\beta = 5A^{-2/3}$. On the prolate side ($\gamma = 0$) and on the oblate side ($\gamma = 60°$) we see again the same spectra as in Fig. 3.10. However, these two limits are now connected through a circle in the β, γ plane (Fig. 1.4) with constant deformation β. We no longer pass through the weak coupling limit at $\beta = 0$. On both sides the spectra do not depend very drastically on the triaxiality γ. The essential transition from the strongly coupled to the decoupled scheme takes place in a relatively small γ region around $\gamma = 30°$. There are many spectra in weakly deformed transitional nuclei that can be nicely reproduced with γ-values between 20° and 40°. This agreement can be improved even more by incorporating the change of the moment of inertia in a VMI-model type calculation [FT 75].

* Recently, Bengtsson et al. [BHM 78] have found an oscillating behavior of this interaction as a function of the chemical potential λ. For an interpretation of this fact see [FPS 80] and the references given there.

† For extension of the triaxial rotor model to multiparticle configurations, see [TNV 77, YTF 77, TYF 77, YTF 78].

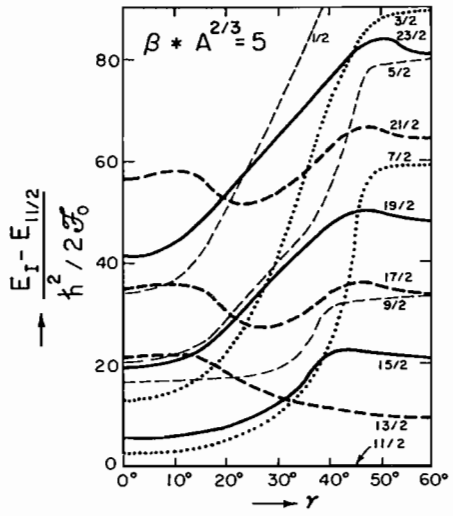


Figure 3.12. Spectrum of a $j=11/2$ particle coupled to an asymmetric rotor with all yrast levels $I<23/2$ as function of γ. (From [MSD 74].)

The *dynamical behavior* of a system of an odd high-j particle coupled to a rotating trixial core is determined by three physical effects (see [Me 75]):

(i) The core prefers to rotate around the axis with the maximal moment of inertia in order to minimize the rotational energy.
(ii) The particle moving in the deformed well prefers maximal mass overlap with the core, because in this case its potential energy is minimal.
(iii) The Coriolis interaction tries to align the angular momenta of the particle **j** and the core **R**.

For high j-values the alignment dominates as long as we have only one particle in the j-shell. Since the density distribution of an aligned particle is oblate with j as symmetry axis, condition (ii) is optimally satisfied if the core is oblate ($\gamma=60°$), which corresponds to the 2-axis as the symmetry axis. However, in this case $\mathcal{I}_2=0$ and condition (i) is violated. Therefore, **R** will be perpendicular to the 2-axis and we no longer have alignment. In the triaxial case, $\mathcal{I}_2$ no longer vanishes and conditions (i), (ii), and (iii) can all be satisfied together. There is an alignment of **j** and **I** along the 2-axis. In fact, the calculations show that the odd particle tends to align along the 2-axis of the core. This axis serves as an approximate symmetry axis in the sense that the approximate quantum numbers $\bar{K}$ and $\bar{\Omega}$ (representing the projections of **I** and **j** onto the 2-axis) are meaningful for a classification. The result is a new level scheme characteristic for a triaxial rotor.

The yrast levels are given by $\bar{\Omega}=j$ and $\bar{K}=j, j\pm2, j\pm4$, as in the rotation aligned coupling scheme of Stephens. For these levels, **I**, **j**, **R**, and the 2-axis are parallel. They lie on the usual parabola with $\Delta I=2$ (see Fig. 3.13). Contrary to the axial symmetric case, where the direction of the alignment was arbitrary in the 1, 2 plane (usually one chooses the 1-axis), we have the asymmetric case with no such symmetry. Alignment favors the 2-axis. This has the consequence that on each yrast level there exists a rotational band with the spin order $\Delta I=1$. In fact, such levels have been experimentally observed [ABR 75].

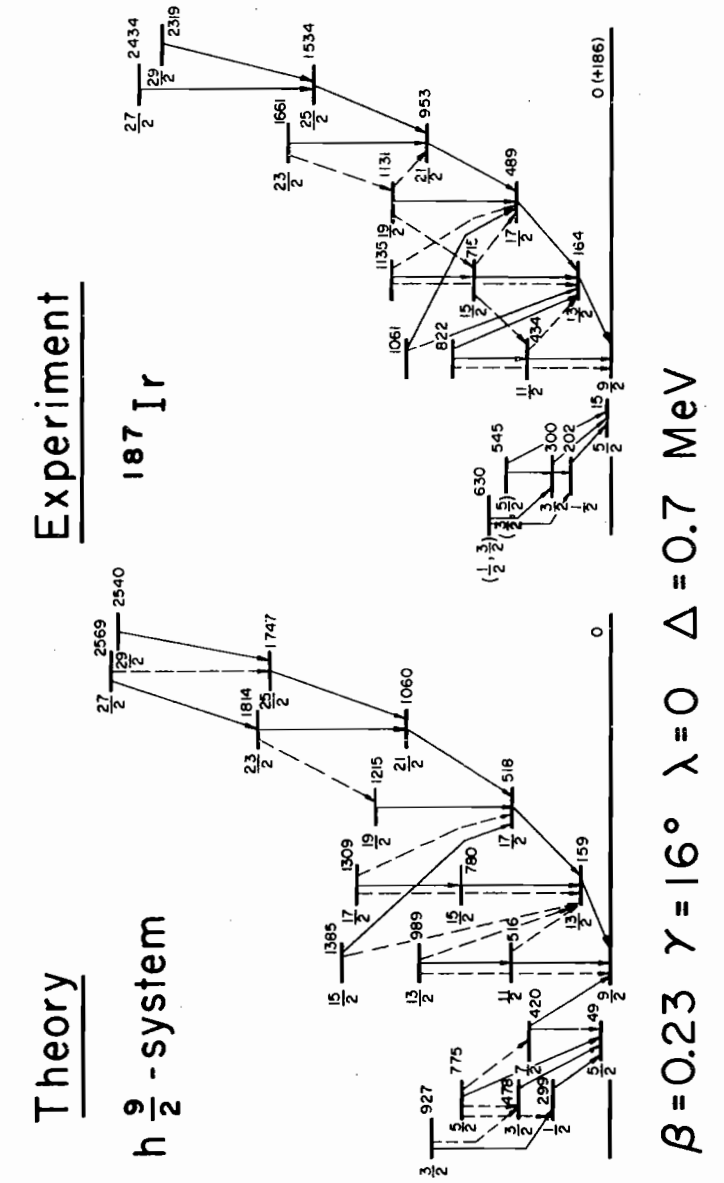


Figure 3.13. The $h_{9/2}$ family of negative parity states in ^{187}Ir. (From [Me 75].)

3.3.4 Electromagnetic Properties

According to the separation of the system into valence particles and a core, we also have two contributions for the electromagnetic multipole operators:

$$Q_{\lambda\mu} = Q_{\lambda\mu}^{(c)} + Q_{\lambda\mu}^{(p)}. \tag{3.60}$$

The core part is given by Eq. (1.35) and $Q^{(p)}$ is defined in Eqs. (B.23, B.24) where the sum over i however is restricted to the number of valence particles.

In the following we shall restrict ourselves to the axially symmetric case without pairing. It is then relatively simple to generalize the formulae to more complicated cases.

The *magnetic dipole moment* of the rotor is given by Eq. (1.36) as $\mu = \mu_N \cdot g_R \mathbf{R}$. We therefore gain for the $M1$ operator:

$$\mathbf{M}_1 = \sqrt{\frac{3}{4\pi}}\ \mu_N \cdot g_R(\mathbf{I}-\mathbf{j}) + \mathbf{M}_1^{(p)}. \tag{3.61}$$

The magnetic dipole moment is defined in Eq. (B.31). With the wave function of Eq. (3.25) we obtain for $K \neq 1/2$, using the projection theorem (2.66),

$$\mu = \sqrt{\frac{4\pi}{3}}\ \langle \Psi_{IK}^{I} | M_{1z} | \Psi_{IK}^{I} \rangle = \mu_N \left\{ g_R \frac{I(I+1)-K^2}{I+1} + g_K \frac{K^2}{I+1} \right\} \tag{3.62}$$

where we have introduced the gryomagnetic ratio g_K in the following way.

$$\begin{aligned} g_K \cdot K &= \frac{1}{\mu_N} \sqrt{\frac{4\pi}{3}}\ \langle \Phi_K | M_{13}^{(p)} | \Phi_K \rangle \\ &= \langle \Phi_K | g_s \cdot s_3 + g_l I_3 | \Phi_K \rangle \\ &= g_l K \cdot + (g_s - g_l) \cdot \langle \Phi_K | s_3 | \Phi_K \rangle. \end{aligned} \tag{3.63}$$

The matrix element $\langle \Phi_K | s_3 | \Phi_K \rangle$ can be calculated with a Nilsson wave function. For $K = 1/2$ one gets an additional contribution to the magnetic moment which contains the decoupling factor a of Eq. (3.30).

The magnetic moments obtained from Eq. (3.62) have been calculated for a large range of nuclei [MN 59, NN 65 p. 653] and one has found that agreement with experimental data is usually better for protons than for neutrons. A detailed investgation shows, however, that one again needs polarization charges for g_s: $g_s^{\text{eff}} \simeq 0.7 g_s^{\text{free}}$ (see [BM 75 p. 302ff]).

Similarly, we can calculate the reduced matrix elements for *$M1$-transitions*. The operator I does not cause any transitions. For the rest we use Eq. (1.70) and get for $K \neq 1/2$ and $I_i \neq I_f$:

$$B(M1, I_i \to I_f) = \frac{1}{2I_i+1} \left| \langle \Psi_K^{I_f} \| M_1 \| \Psi_K^{I_i} \rangle \right|^2 = \frac{3}{4\pi} \mu_N^2 (g_K - g_R)^2 K^2 \left| C_{K\ 0\ K}^{I_i\ 1\ I_f} \right|^2. \tag{3.64}$$

These $M1$-transitions are only possible for rotational bands in odd nuclei. Also, reasonable agreement with experiment has been found [SBP 67].

Again using Eq. (1.70) we can also derive the $B(E2)$ values for *electric quadrupole transitions*. For transitions between states with the same K value the core contribution is much larger than the particle contribution because of the large Q_0 values. Therefore, the latter can be neglected, and we obtain

$$B(E2, I_i \to I_f) = Q_0^2 \frac{5}{16\pi} \left| C_{K\ 0\ K}^{I_i\ 2\ I_f} \right|^2. \tag{3.65}$$

This equation is used to determine the experimental values for the deformations appearing in Q_0. In Sec. 2.8.6 we saw that they agree with the theoretically determined deformations. Therefore, the particle-plus-rotor model gives the proper $E2$ transition probabilities for transitions with the same K-value.

For transitions with different K-values, the collective part vanishes. Such transitions are, in fact, very weak (*K-forbidden*) because the single-particle part contains only the single-particle matrix element of r^2Y_{20} in the intrinsic frame. We have already seen in Section 2.7.2 that they are small compared to the collective values. On the other hand, a pure single-particle model cannot explain the effective charges in spherical nuclei (see Sec. 2.7.2). The same difficulties occur here again. For a detailed discussion, see the paper of Löbner and Malinskog [LM 66], which contains much experimental data together with possible ways to improve the simple Nilsson estimate.

Finally, we have to mention that the above considerations apply only to pure K-bands. For transition probabilities and electromagnetic moments in K-mixed bands, like rotational aligned bands or bands in an asymmetric rotor, we have to take into account the mixing coefficients.

3.4 The Cranking Model*

We have seen in the last section how the motion of particles in a deformed well can be connected with the idea of a rigid rotor. This model is very successful in the description of the level structure of rotational and even transitional nuclei. However there exists no straightforward microscopic derivation; in particular, one cannot calculate the inertial parameters in this model.

On the other hand, nearly all fully microscopic theories of nuclear rotation are based on or related in some way to the cranking model, which was introduced by Inglis [In 54, 56] in a semiclassical way, but as we shall see in Section 11.4, it can be derived fully quantum mechanically, at least in the limit of large deformations, and not too strong K-admixtures ($K \ll I$).

The cranking model has the following advantages.

(i) In principle, it provides a fully microscopic description of the rotating nucleus. There is no introduction of redundant variables, therefore, we are able to calculate the rotational inertial parameters microscopically within this model and get a deeper insight into the *dynamics* of rotational motion.

(ii) It describes the collective angular momentum as a sum of single-particle angular momenta. Therefore, collective rotation as well as single-particle rotation, and all transitions in between such as decoupling processes, are handled on the same footing.

(iii) It is correct for very large angular momenta, where classical arguments apply (even if the quantum mechanical derivation does not work in this limit [BMR 70]).

* In this chapter, we treat only cranking theory for rotations. We can, however, also apply a similar theory for general collective motions, as discussed in Section 12.3.7.

The shortcomings of the model are:

(i) As we shall see, it is basically a nonlinear theory. Only in the limit of small angular momenta can one linearize it using perturbation theory (cranking formula for the moment of inertia). In general, the calculations are therefore complicated, especially in cases where one has several solutions.

(ii) The resulting wave functions are not eigenstates of the angular momentum operators. It is therefore not clear a priori how one has to calculate, for example, electromagnetic transition probabilities. In fact, we shall see in Section 11.4 that cranking model wave functions are in a sense only *internal wave functions* and that one has to use projection techniques to get the wave functions in the laboratory system.

In the following we shall give the usual semiclassical derivation (see, for instance, [Vi 57b, So 73]) and discuss the cranking model in connection with a pure single-particle Hamiltonian. Many of the arguments in the next sections can, however, also be applied to a general two-body Hamiltonian (see Sec. 7.7).

3.4.1 Semiclassical Derivation of the Cranking Model

The basic idea of the cranking model is the following classical assumption: If one introduces a coordinate system which rotates with constant angular velocity ω around a fixed axis in space, the motion of the nucleons in the rotating frame is rather simple if the angular frequency is properly chosen; in particular, the nucleons can be thought of as independent particles moving in an average potential well which is rotating with the coordinate frame.

In Section 11.4, we will see how the consequences of this picture can also be derived from quantum mechanics using projection techniques. For the moment, however, we want to stay with the classical description because of its intuitive character. Also, we do not want to take into account any residual interaction. Therefore, we assume a single-particle potential V of fixed shape, which rotates in space, and accordingly we must consider the time-dependent single-particle Hamiltonian

$$h(t)=\frac{\mathbf{p}^2}{2m}+V(\mathbf{r},t) \tag{3.66}$$

and the corresponding Schrödinger equation

$$h(t)\psi(t)=i\hbar\frac{\partial}{\partial t}\psi(t). \tag{3.67}$$

Introducing spherical coordinates r, θ, φ with respect to the axis ω, we can represent the time-dependence of $V(t)$ in the following way. If $V(\mathbf{r}, 0)$ is the

potential at time $t=0$, then we have at time t:

$$V(\mathbf{r},t)=V(r,\theta,\varphi-\omega t,0). \tag{3.68}$$

Again we realize that V is only time dependent if it depends on φ. In other words, V should not have axial symmetry around the rotational axis, because then there can be no collective rotation possible around an axis of symmetry for a quantum mechanical system (see Sec. 1.5.1). Because of the very simple time dependence of $V(t)$ in Eq. (3.68), a unitary transformation exists which eliminates this time dependence. It is

$$U=e^{i\omega \mathbf{l}t} \tag{3.69}$$

with $\omega\cdot\mathbf{l}=(\hbar/i)\omega\cdot\partial/\partial\varphi$. U produces a rotation of an angle ωt around the rotational axis ω.

We find the time-dependent operator

$$Uh(t)U^{-1}=h(0) \tag{3.70}$$

and define

$$\tilde{\psi}=U\psi, \tag{3.71}$$

with

$$i\hbar\dot{\tilde{\psi}}=i\hbar U\dot{\psi}+i\hbar\dot{U}\Psi=(h(0)-\omega\mathbf{l})\tilde{\psi}. \tag{3.72}$$

Equation (3.72) is a time-dependent Schrödinger equation with an explicitly time-independent effective Hamiltonian h_ω. It thus can be solved as an eigenvalue problem in the standard way:

$$h_\omega\tilde{\psi}=(h(0)-\omega\mathbf{l})\tilde{\psi}=\epsilon'_\omega\tilde{\psi}, \tag{3.73}$$

where ϵ'_ω are eigenvalues of the effective Hamiltonian. To get the energies of the original Hamiltonian, we have to calculate

$$\epsilon_\omega=\langle\psi|h(t)|\psi\rangle=\langle\tilde{\psi}|h(0)|\tilde{\psi}\rangle=\epsilon'_\omega+\omega\langle\tilde{\psi}|\mathbf{l}|\tilde{\psi}\rangle. \tag{3.74}$$

The term $\omega\mathbf{l}$ is usually called the Coriolis term.

We have now solved the time-dependent Schrödinger equation in a rotating potential and found that we must diagonalize an effective time-independent Hamiltonian. We want to emphasize that we have not derived a priori the Hamiltonian as it is seen from the rotating coordinate system, since we transformed only the coordinates and not the momenta. In fact, in the case of pure translational motion, we would get a similar result [$h_v=h(0)-\mathbf{v}\cdot\mathbf{p}$], but from Galilean invariance we require that the Hamiltonian seen from the moving coordinate system is the same as in the rest frame. Nevertheless, it turns out that in the special case of rotations, h_ω in Eq. (3.73) is identical with the Hamiltonian as seen from the rotating system [Va 56; Br 64, p. 69]. From the term $\omega\mathbf{l}$, we can derive the Coriolis force as well as the centrifugal force.

For systems with spin the operator which generates rotations is $\mathbf{j}=\mathbf{l}+\mathbf{s}$. The orientation of the rotational axis is usually chosen as parallel to the x-axis because it is understood to be perpendicular to the axis of symme-

try, which for $\omega=0$ is the z-axis. For higher angular momenta, one also investigates nonsymmetric single-particle potentials. Nevertheless, we require that ω is parallel to a principal axis of the potential. Therefore, the many-body Hamiltonian of the cranking model is given by ($J_x=\sum_{i=1}^{A} j_x^{(i)}$):

$$H_\omega = H-\omega J_x = \sum_{i=1}^{A} h_\omega^{(i)}, \tag{3.75}$$

where H is a sum of deformed single-particle Hamiltonians.

Within the cranking model we must now diagonalize H_ω, and the resulting ground state wave function Φ_ω is a Slater determinant. As in the normal shell model (with $\omega=0$) the question arises how the levels in the single-particle potential should be filled to obtain the lowest energy state for any given angular momentum (the yrast level). The answer to this question (given in Sec. 11.4) is that we have to minimize the energy $E'=\langle\Phi|H-\omega J_x|\Phi\rangle$, that is, we have to fill up the potential in the usual way *in the rotating frame*.

For the energy in the laboratory system, from Eq. (3.74) we get

$$E(\omega)=\langle\Phi_\omega|H|\Phi_\omega\rangle=\langle\Phi_\omega|H_\omega|\Phi_\omega\rangle+\omega\langle\Phi_\omega|J_x|\Phi_\omega\rangle. \tag{3.76}$$

Since $E(\omega)$ cannot depend on the sign of ω, one finds

$$E(\omega)=E(0)+\frac{1}{2}\mathcal{I}_1\omega^2+\cdots \tag{3.77}$$

and, since for $\omega=0$ $\langle\Phi_0|J_x|\Phi_0\rangle=0$,

$$J(\omega)=\langle\Phi_\omega|J_x|\Phi_\omega\rangle=\mathcal{I}_2\omega+\cdots. \tag{3.78}$$

We can show that the constants $\mathcal{I}_1$ and $\mathcal{I}_2$ are equal [Sch 61],

$$\mathcal{I}_1=\mathcal{I}_2, \tag{3.79}$$

using the fact that $E(\omega)$ is the lowest eigenvalue of H_ω. According to the variation principle of Ritz, we get Φ_ω as a solution of the equation

$$\delta\langle\Phi|H-\omega J_x|\Phi\rangle=0, \tag{3.80}$$

where $|\Phi\rangle$ is any one from the family of all possible Slater determinants. The condition is also fulfilled if we take $|\Phi\rangle$ to be taken out of the set $\{\Phi_{\omega'}\}$, where $\Phi_{\omega'}$ is an eigenfunction of $H-\omega' J_x$, and ω' runs through all real numbers. Then we find from Eq. (3.80):

$$\frac{d}{d\omega'}\langle\Phi_{\omega'}|H|\Phi_{\omega'}\rangle-\omega\frac{d}{d\omega'}\langle\Phi_{\omega'}|J_x|\Phi_{\omega'}\rangle=0 \tag{3.81}$$

or

$$\mathcal{I}_1=\frac{1}{\omega}\frac{d}{d\omega}E(\omega)\Big|_{\omega=0}=\frac{d}{d\omega}J(\omega)\Big|_{\omega=0}=\mathcal{I}_2.$$

We also derive from Eq. (3.80);

$$\omega=\frac{dE}{dJ}. \tag{3.82}$$

To have a comparison with experiment, we have to determine the value of

the angular velocity. Inglis [In 54] proposed to include the zero-point oscillations at least semiclassically by requiring

$$J=\langle\Phi_\omega|J_x|\Phi_\omega\rangle=\sqrt{I(I+1)}\ . \tag{3.83}$$

In first order we get

$$\omega=\frac{\sqrt{I(I+1)}}{\vartheta_1} \tag{3.84}$$

and, from (3.77),

$$E(I)=E(0)+\frac{1}{2\vartheta_1}I\cdot(I+1). \tag{3.85}$$

For higher ω-values there are deviations from this $I(I+1)$ law. In general, the moment of inertia is defined as

$$\vartheta=\frac{J}{\omega}\ . \tag{3.86}$$

Up to now we have investigated only completely independent particle motion, by which we mean that we have even neglected the influence of the rotation on the average field. In Section 7.7 we will show how this can be taken into account.

3.4.2 The Cranking Formula

In the case of a pure $I(I+1)$ spectrum, we need calculate only one constant, the *moment of inertia*. It is already determined by the 2^+ state and therefore it seems meaningful to apply perturbation theory for such small I-values.

We start with the unperturbed system of a deformed potential, which is filled up to the Fermi level. Levels below will be called holes (indices $i, i', \ldots$); levels above will be called particles (indices $m, m', \ldots$). The shell model basis consists of the ground state $|\Phi_0\rangle$, *ph*-states $|mi\rangle=a_m^+a_i|\Phi_0\rangle$, $2p$-$2h$ states, and so on. The perturbation $\omega\cdot J_x$ is a one-particle operator and can therefore excite only one *ph* pair at a time. Therefore, we get for the perturbed wave function up to first order

$$|\Phi\rangle=|\Phi_0\rangle+\omega\sum_{im}\frac{\langle mi|J_x|\Phi_0\rangle}{\epsilon_m-\epsilon_i}a_m^+a_i|\Phi_0\rangle, \tag{3.87}$$

where ϵ_i and ϵ_m are the single-particle energies of the Hamiltonian H. The expectation value of J_x up to first order in ω is then

$$J=\langle\Phi|J_x|\Phi\rangle=2\omega\sum_{mi}\frac{|\langle mi|J_x|\Phi_0\rangle|^2}{\epsilon_m-\epsilon_i}\ , \tag{3.88}$$

which, together with (3.78) gives for the moment of inertia*

$$\vartheta_{\text{Inglis}}=2\cdot\sum_{mi}\frac{|\langle m|J_x|i\rangle|^2}{\epsilon_m-\epsilon_i}\ . \tag{3.89}$$

*In molecular physics a similar formula has been derived by Wick [Wi 48].

This is the well known *Inglis formula* for the moment of inertia [In 54, BM 55].

The moments of inertia that result from this formula are usually very close to the rigid body value of the moment of inertia [Eq. (1.49)]. In fact, we shall see in Section 3.4.3 that, in the case of a pure anisotropic oscillator, this will be an exact result. Lüders [Lü 60] showed that this is the result for any independent particle model in the limit of large particle numbers (see also [AB 59, Ro 59, SB 64, Da 75, KG 78]).

We can understand this result qualitatively, if we realize that the velocity distribution of the ground state in a deformed static potential is nearly isotropic (see Sec. 13.3), and that this fact is not changed in the rotating system by Coriolis or centrifugal forces (see [BJ 76b]). Then there is no net flow in the intrinsic system and from the laboratory frame we observe a rigid-rotation velocity distribution.

As we have seen in Section 1.5.1, the experimental moments of inertia are a factor of 2 to 3 smaller than their rigid body values. Bohr and Mottelson [BM 55, Mo 56] already indicated that residual two-body interactions would lower these values. The most important influences in this respect are the correlations of the pairing type. Since they can be included very easily within the BCS-formalism (see Chap. 6) in a single-particle description, we give here the derivation of the so-called Belyaev formula [Be 59, 61], which is the extension of the Inglis formula (3.89) that includes pairing correlations. (Readers not familiar with this formalism are referred to Chapter 6.)

In this case, $|\mathrm{BCS}\rangle$ represents the BCS-ground state (6.31) and excitations are given by the two-quasi-particle states $\alpha_k^+\alpha_{k'}^+|\mathrm{BCS}\rangle$, four-quasi-particle states, etc. By analogy with Eq. (3.87), we obtain the perturbed wave function

$$|\Phi\rangle = |\mathrm{BCS}\rangle + \omega \sum_{k<k'} \frac{\langle \mathrm{BCS}|\alpha_{k'}\alpha_k J_x|\mathrm{BCS}\rangle}{E_k + E_{k'}} \alpha_k^+\alpha_{k'}^+|\mathrm{BCS}\rangle, \tag{3.90}$$

where $E_k + E_{k'}$ is the excitation energy of the quasi-particle pair k, k'. The quasi-particle energies are given by

$$E_k = \sqrt{(\epsilon_k - \lambda)^2 + \Delta_k^2}\ . \tag{3.91}$$

Proceeding as in Eqs. (3.88) and (3.89), we find for the moment of inertia

$$\mathcal{I}_{\mathrm{Belyaev}} = 2 \sum_{k<k'} \frac{|J^{20}_{x_{kk'}}|^2}{E_k + E_{k'}}\ . \tag{3.92}$$

From Eq. (E.16), we find J_x^{20} and get

$$\mathcal{I}_{\mathrm{Belyaev}} = 2 \sum_{k,\,k'>0} \frac{|\langle k|J_x|k'\rangle|^2}{E_k + E_{k'}} (u_k v_{k'} - u_{k'} v_k)^2. \tag{3.93}$$

This formula for the moment of inertia indeed yields lower values as compared to expression (3.89). Two effects are responsible for this:

(i) The energy denominator is much larger than the particle-hole energies in Eq. (3.89). The parameter Δ [Eq. (3.91)] produces a gap of at least $2\Delta \simeq 2$ MeV for the important levels in the neighborhood of the Fermi surface

(ii) The factor $(u_k v_{k'} - u_{k'} v_k)^2$ is usually somewhat smaller than unity.

The lowering of the moment of inertia in the BCS-theory, as compared to the rigid body value, corresponds to a superfluid slippage of some nucleons as the nucleus rotates [Mi 59, 60].

Extended numerical calculations [GR 60, NP 61, MT 75] for realistic nuclei show a remarkable agreement with the experimental values. As we shall see in the next section, it is very important to apply the Inglis or the Belyaev formula to self-consistent wave functions, that is, those calculated at deformations that correspond to the energy minimum. In most of these calculations, this is achieved by using Nilsson wave functions and energies at the experimentally observed deformations as well as the experimentally determined values of the gap Δ (see also [MN 59]).

The success of these calculations, which produce roughly correct moments of inertia lying between the (too small) irrotational values and the (too large) rigid body value gives us great confidence that the picture of rotational nuclei as a deformed superfluid many-body system is correct.

Of course, we can investigate the influence of the *residual interaction* on the moment of inertia. This can be done within the framework of linear response theory (see Sec. 8.5.3 and [MSV 72]). In this kind of theory, the external field represented by the Coriolis operator J_x can excite virtual vibrations of the core which, in turn, have an influence on the moment of inertia. There are two types of such vibrations: surface oscillations (*ph* vibrations), which correspond to the stretching effect, and oscillations in the pairing correlations (*pp*-vibrations; see Sec. 8.3.5). The net result of such calculations is that both effects more or less cancel, and we get roughly the same values for the moments of inertia (Fig. 3.14) as given by the BCS theory.

We have now discussed the application of perturbation theory to the calculation of the expectation value of J_x, that is, the moment of inertia. In a similar fashion, we can calculate other properties of the rotating nucleus, for example, the *gyromagnetic ratio* g_R or the magnetic moment of the first 2^+ state. Since the magnetic moment μ is defined as the expectation value of μ_z in the state $|I, M=I\rangle$ [Eq. (B.31)] and the cranking model wave functions are not eigenfunctions of angular momentum, it is not clear at this point how to calculate μ. In Sec. 11.4, we will see that a projection technique has to be applied. In lowest order, we get a very simple result, which can be understood easily within the semiclassical picture of the cranking model:

$$\mu = \langle \Phi_\omega | \mu_x | \Phi_\omega \rangle. \tag{3.94}$$

We can therefore define the gyromagnetic ratio (1.37) by

$$\mu = g_R \cdot J = g_R \cdot \langle \Phi_\omega | J_x | \Phi_\omega \rangle. \tag{3.95}$$

From (3.90), in first order perturbation theory, we get

$$g_R = \frac{1}{\mathcal{I}} \sum_{k,k'>0} \frac{(\langle k|J_x|k'\rangle\langle k'|\mu_x|k\rangle + \text{c.c.})}{E_k + E_{k'}} (u_k v_{k'} - u_{k'} v_k)^2. \tag{3.96}$$

The values calculated with these formulae are smaller than the liquid drop value $g_R = Z/A$ [Eq. (1.37)] and agree quite well with the experimental data [MSV 72].

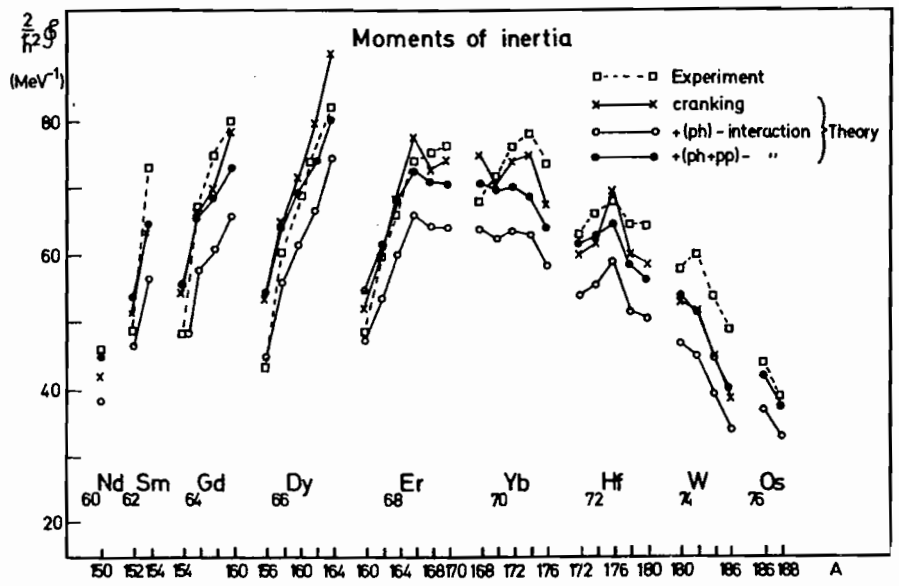


Figure 3.14. Moments of inertia in rare earth nuclei. The squares indicate experimental values [NP 61]; the crosses are obtained from the Belyaev formula (3.93). The open circles take into account only the stretching effect (*ph* interaction); in addition to that, the closed circles also take into account the antipairing effect (*pp*-interaction). (From [MSV 72].)

As we have seen in Section 3.2, deviations from the $I(I+1)$ law occur as we go to higher angular momenta. To calculate the *B and C coefficients* [Eq. (3.1)] connected with these deviations, one has used perturbation theory in higher order, including the effects of a residual interaction [Ma 65, Ma 67a, MR 70], so that the most important effect is the so-called Coriolis-antipairing effect, which we will discuss in Section 7.7. As we shall see, it is only at the very high spin states with $I > 30$ or 40 that one can expect the pairing correlations to vanish.

3.4.3 The Rotating Anisotropic Harmonic Oscillator

We have seen in Section 2.8.3 that the anisotropic harmonic oscillator can be solved analytically and that it provides at least a qualitative model for a deformed nucleus. It turns out that it can also be solved analytically in the rotating frame (see [Va 56, Ze 75, RBK 75, GMZ 78]), because $l_x = yp_z - zp_y$ is a quadractic form in the coordinates x, y, z and momenta p_x, p_y, p_z.

We can forget about the spin, since the potential does not depend on it (the term $-\omega s_x$ in the Coriolis operator can be diagonalized separately in the spin space and gives only a diagonal contribution). The Hamiltonian then has the form

$$h_\omega = -\frac{\hbar^2}{2m}\Delta + \frac{1}{2}m\left(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2\right) - \omega(yp_z - zp_y). \tag{3.97}$$

We introduce the creation and annihilation operators for the harmonic oscillator

bosons in x, y, and z directions a_x, a_y, and a_z:

$$\begin{aligned} x &= \left(\frac{\hbar}{2m\omega_x}\right)^{\frac{1}{2}}(a_x + a_x^+) \qquad p_x = \frac{\hbar}{i}\left(\frac{m\omega_x}{2\hbar}\right)^{\frac{1}{2}}(a_x - a_x^+), \\ y &= i\left(\frac{\hbar}{2m\omega_y}\right)^{\frac{1}{2}}(a_y - a_y^+) \qquad p_y = \hbar\left(\frac{m\omega_y}{2\hbar}\right)^{\frac{1}{2}}(a_y + a_y^+), \\ z &= \left(\frac{\hbar}{2m\omega_z}\right)^{\frac{1}{2}}(a_z + a_z^+) \qquad p_z = \frac{\hbar}{i}\left(\frac{m\omega_z}{2\hbar}\right)^{\frac{1}{2}}(a_z - a_z^+). \end{aligned} \tag{3.98}$$

In these operators, h_ω is of the form

$$\begin{aligned} h_\omega = \hbar\omega_x\left(a_x^+ a_x + \frac{1}{2}\right) + \hbar\omega_y\left(a_y^+ a_y + \frac{1}{2}\right) + \hbar\omega_z\left(a_z^+ a_z + \frac{1}{2}\right) \\ + \frac{\hbar\omega}{2\sqrt{\omega_y\omega_z}}\{(\omega_y - \omega_z)(a_y^+ a_z^+ + a_y a_z) + (\omega_y + \omega_z)(a_y^+ a_z + a_z^+ a_y)\}. \end{aligned} \tag{3.99}$$

The Coriolis operator in (3.99) has two contributions. The first one creates or annihilates two oscillator quanta. Therefore, it couples shells with major quantum numbers N and $N \mp 2$.* The second part conserves the total number of quanta, but shifts quanta from the y-direction into the z-direction and vice versa.

3.4.3.1 The Inglis Formula. First we want to treat the Coriolis term in perturbation theory. In the Inglis formula (3.89), the $\Delta N = \pm 2$ part has the energy denominator $\pm\hbar(\omega_y + \omega_z)$ and the $\Delta N = 0$ part has the denominator $\pm\hbar(\omega_y - \omega_z)$. The matrix elements of a, a^+ are given in Eq. (C. 12f),

$$\langle n-1|a|n\rangle = \sqrt{n}\ , \qquad \langle n+1|a^+|n\rangle = \sqrt{n+1}\ ,$$

and we get, from (3.89)

$$\mathcal{I}_{\text{Inglis}} = \frac{\hbar}{2\omega_y\omega_z}\left\{\frac{(\omega_y - \omega_z)^2}{\omega_y + \omega_z}(N_y + N_z) + \frac{(\omega_y + \omega_z)^2}{\omega_y - \omega_z}(N_z - N_y)\right\}, \tag{3.100}$$

where the

$$N_{x,\,y,\,z} = \sum_{i=1}^{A}\left(n_{x,\,y,\,z} + \frac{1}{2}\right)_i \tag{3.101}$$

satisfy the relations

$$\bar{x}^2 = \frac{1}{A}\sum_{i=1}^{A}\langle x^2\rangle_i = \frac{1}{A}\frac{\hbar}{m\omega_x}\sum_{i=1}^{A}\left(n_x + \frac{1}{2}\right)_i = \frac{\hbar}{Am\omega_x}\cdot N_x. \tag{3.102}$$

Very important for the following discussion is the *self-consistency condition*, [BM 75], namely, that the potential (given by $\omega_x, \omega_y, \omega_z$) has the same shape as the density distribution (given by $\bar{x}^2$, $\bar{y}^2$ and $\bar{z}^2$):

$$\sqrt{\bar{x}^2}\ :\sqrt{\bar{y}^2}\ :\sqrt{\bar{z}^2} = \frac{1}{\omega_x} : \frac{1}{\omega_y} : \frac{1}{\omega_z}. \tag{3.103}$$

Together with Eq. (3.102), we find

$$\omega_x N_x = \omega_y N_y = \omega_z N_z = C. \tag{3.104}$$

* It vanishes for small deformations ($\omega_y \simeq \omega_z$) and is therefore often neglected [ALL 76].

With this condition we can rewrite Eq. (3.100):

$$\mathcal{I}_{\text{Inglis}} = \hbar \left\{ \frac{N_y}{\omega_y} + \frac{N_z}{\omega_z} \right\}. \tag{3.105}$$

The value of the moment of inertia so obtained is identical to the rigid body value. From Eq. (3.102) we get

$$\mathcal{I}^{\text{rig}} = m \sum_{i=1}^{A} \left(\langle y^2 \rangle_i + \langle z^2 \rangle_i \right) = \hbar \left(\frac{N_y}{\omega_y} + \frac{N_z}{\omega_z} \right). \tag{3.106}$$

Once again, we would like to stress the point that the self-consistency condition (3.104) is crucial for this result. If we occupied the y- and z-direction with the same number of oscillator quanta ($N_y = N_z$), we would get from Eqs. (3.100) and (3.102) a value

$$\mathcal{I} = \hbar \frac{(N_y/\omega_y - N_z/\omega_z)^2}{(N_y/\omega_y + N_z/\omega_z)} = Am \frac{(\bar{y}^2 - \bar{z}^2)^2}{\bar{y}^2 + \bar{z}^2}, \tag{3.107}$$

which is proportional to the irrotational flow value [BM 75].

3.4.3.2 Exact Solution. In the next step, we go beyond perturbation theory and diagonalize the single-particle Hamiltonian (3.99) exactly.* It is a quadratic form in the boson operators a, a^+ and can therefore be diagonalized [RBK 75] in the same way as one diagonalizes quadratic forms of the RPA-type (see Chap. 8).

To diagonalize the single-particle Hamiltonian, we introduce a canonical transformation among the momenta p_y, p_z and the coordinates y, z [Va 56]:

$$\begin{aligned} Q_2 &= \alpha_2 (y + \beta p_z), \\ Q_3 &= \alpha_3 (z + \beta p_y), \\ P_2 &= \alpha_2^{-1} (1 - \delta\beta)^{-1} (p_y + \delta z), \\ P_3 &= \alpha_3^{-1} (1 - \delta\beta)^{-1} (p_z + \delta y), \end{aligned} \tag{3.108}$$

which guarantees that the Q_i, P_i fulfill the commutation relations of momenta and coordinates. The constants β and δ are determined by the requirement that the Hamiltonian (3.97) contains no mixed terms $P_2 Q_3$ or $P_3 Q_2$ in the new representation. The constants α_2 and α_3 normalize the new coordinates in such a way that the mass parameter is again m. In the new variables, the Hamiltonian has the form [Va 56]

$$h_\omega = \left(-\frac{p_x^2}{2m} + \frac{1}{2} m \omega_x^2 x^2 \right) + \left(-\frac{P_2^2}{2m} + \frac{1}{2} m \Omega_2^2 Q_2^2 \right) + \left(-\frac{P_3^2}{2m} + \frac{1}{2} m \Omega_3^2 Q_3^2 \right). \tag{3.109}$$

The coordinates x, Q_2, Q_3 are the normal coordinates of the problem, and the frequencies Ω_i are given by

$$\Omega_{2,3}^2 = \omega^2 + \omega_+^2 \pm \sqrt{\omega_-^4 + 4\omega^2 \omega_+^2}, \tag{3.110}$$

with

$$\omega_\pm^2 = \frac{1}{2} (\omega_y^2 \pm \omega_z^2). \tag{3.111}$$

* For the calculation of matrix elements in this rotating oscillator basis, see [LR 77].

We can now define rotating bosons B_2^+, B_3^+ in analogy to Eq. (3.98):

$$B_2^+ = (2\hbar m\Omega_2)^{-\frac{1}{2}}(P_2 + im\Omega_2 Q_2), \tag{3.112}$$
$$B_3^+ = -i(2\hbar m\Omega_3)^{-\frac{1}{2}}(P_3 + im\Omega_3 Q_3),$$

and obtain for the Hamiltonian

$$h_\omega = \hbar\omega_x\left(a_x^+ a_x + \frac{1}{2}\right) + \hbar\Omega_2\left(B_2^+ B_2 + \frac{1}{2}\right) + \hbar\Omega_3\left(B_3^+ B_3 + \frac{1}{2}\right). \tag{3.113}$$

The corresponding eigenstates

$$|n_x, n_2, n_3\rangle \propto (a_x^+)^{n_x}(B_2^+)^{n_2}(B_3^+)^{n_3}|-\rangle \tag{3.114}$$

are characterized by the numbers of rotating bosons.

Their single-particle energies are

$$\epsilon'_{n_x n_2 n_3} = \hbar\omega_x\left(n_x + \frac{1}{2}\right) + \hbar\Omega_2\left(n_2 + \frac{1}{2}\right) + \hbar\Omega_3\left(n_3 + \frac{1}{2}\right). \tag{3.115}$$

Assuming again a fixed occupation in a Slater determinant $|\ \rangle$, that is, a fixed set of numbers N_x, N_2, N_3 defined in analogy to Eq. (3.101) for rotating bosons, we find for the total energy in the rotating frame,

$$E'(\omega) = \langle H_\omega\rangle = N_x\hbar\omega_x + N_2\hbar\Omega_2 + N_3\hbar\Omega_3. \tag{3.116}$$

Remembering that $\langle H_\omega\rangle$ is stationary with respect to variations of the eigenfunctions, we can calculate the expectation value of the angular momentum [RBK 75].

$$\langle L_x\rangle = -\langle\frac{\partial H_\omega}{\partial\omega}\rangle = -\frac{\partial}{\partial\omega}E' = \omega\left\{\frac{4\omega_+^2}{\Omega_2^2-\Omega_3^2}\left(\frac{N_3}{\Omega_3}-\frac{N_2}{\Omega_2}\right) - \left(\frac{N_2}{\Omega_2}+\frac{N_3}{\Omega_3}\right)\right\} \tag{3.117}$$

and the shape parameters $\bar{x}^2$, $\bar{y}^2$, $\bar{z}^2$,

$$\bar{x}^2 = \langle x^2\rangle = \frac{1}{m\omega_x}\langle\frac{\partial H_\omega}{\partial\omega_x}\rangle = \frac{1}{m\omega_x}\frac{\partial E'}{\partial\omega_x}, \qquad \bar{y}^2 = \cdots, \tag{3.118}$$

and so on. The moment of inertia is given by

$$\mathcal{I} = \frac{\langle L_x\rangle}{\omega} = m\langle y^2 + z^2\rangle + \frac{4\hbar}{\Omega_2^2-\Omega_3^2}(N_3\Omega_3 - N_2\Omega_2). \tag{3.119}$$

For the final construction of the many-body Slater determinant, however, we have to fix the occupation numbers N_x, N_2, N_3.

In the nonrotating case, the numbers N_x, N_y, N_z were determined by the self-consistency condition (3.104). It can be motivated by different arguments, which all give the same results at $\omega = 0$. For $\omega \neq 0$ this is no longer the case. Several methods have been proposed:

(i) *Minimizing the expectation value* $E'(\omega)$ in Eq. (3.116) for fixed occupation as a function of the deformation parameters ω_x, ω_y and ω_z, and the frequency ω under the constraint of constant volume ($\bar{x}^2\cdot\bar{y}^2\cdot\bar{z}^2 =$ const.) and fixed angular momentum $\langle L_x\rangle$ [St 78, TA 79].

(ii) Requiring an *isotropic velocity distribution* in the rotating frame [RBK 75]. This is reasonable for heavy nuclei, where there are many level crossings and one has always filled up the lowest levels in the potential (see Sec. 13.3). We then get the modified self-consistency condition

$$N_x\omega_x = N_2\Omega_2 = N_3\Omega_3. \tag{3.120}$$

It can be shown that under this condition the shape of the mass distribution (given by $\bar{x}^2, \bar{y}^2, \bar{z}^2$) is proportional to the shape of the potential if one includes the centrifugal potential $\frac{1}{2}m\,(\omega\times\mathbf{r})^2$.

From Eq. (3.119) we see that the self-consistency condition (3.120) yields the rigid-body value for the moment of inertia* at the actual deformation, which may change for large I-values.

3.4.4 The Rotating Nilsson Scheme

For realistic heavy nuclei, the pure harmonic oscillator is only of a limited importance, because it does not contain the drastic energy shift of high j shell orbitals due to the $\mathbf{l}\cdot\mathbf{s}$ term. As we have seen in Section 3.3, they play a crucial role in the interpretation of rotational spectra in all heavy nuclei.

One therefore has extended the Nilsson model and added a Coriolis term $-\omega j_x$ to the single-particle Hamiltonian (2.89) [ALL 76, RNS 78]:

$$h'(\omega)=h-\omega j_x \tag{3.121}$$

Figure 3.15 shows the qualitative behavior of some of the single-particle levels thus obtained as a function of ω. It shows the following features.

(i) At $\omega=0$ are the usual Nilsson levels. They are twofold degenerate with respect to time reversal symmetry ($\pm\Omega$). For $\omega\neq 0$, this symmetry is broken by the Coriolis term, and a split into two single levels is observed.

(ii) The cranked Nilsson Hamiltonian is still invariant under a rotation of 180° around the x-axis, that is, the two levels belong to eigenstates of the operator

$$\mathcal{R}_x = e^{i\pi j_x} \tag{3.122}$$

with the eigenvalues $r_x=\pm i$ ("*signature*" [Bo 76a, b]).

(iii) Some levels show an extremely strong level splitting with increasing ω. They belong to orbits with large j- and small Ω-values (e.g., $1i_{13/2}$, $660\frac{1}{2}$, $651\frac{3}{2}$, $642\frac{5}{2}$). They show strong Ω-mixing and alignment along the x-axis (decoupled bands).

(iv) For even nuclei at moderate angular velocities, however, pairing correlations, as discussed in Section 3.2.2. and 7.7, should be taken into account. They counteract the rotational alignment and try to keep the particles in pairwise occupied orbits. In a full microscopic description of the backbending effect, the pair correlations must be taken into account self-consistently (see Sec. 7.7).

(v) For large frequencies, the alignment effect brings levels from higher major shells down into the neighborhood of the Fermi surface.

*It has recently been shown [FR 76b] that electrons confined by a harmonic potential and submitted to a constant magnetic field are rotating uniformly around the axis formed by the field; this is the analogous effect to the rigid body value of the moment of inertia discussed here.

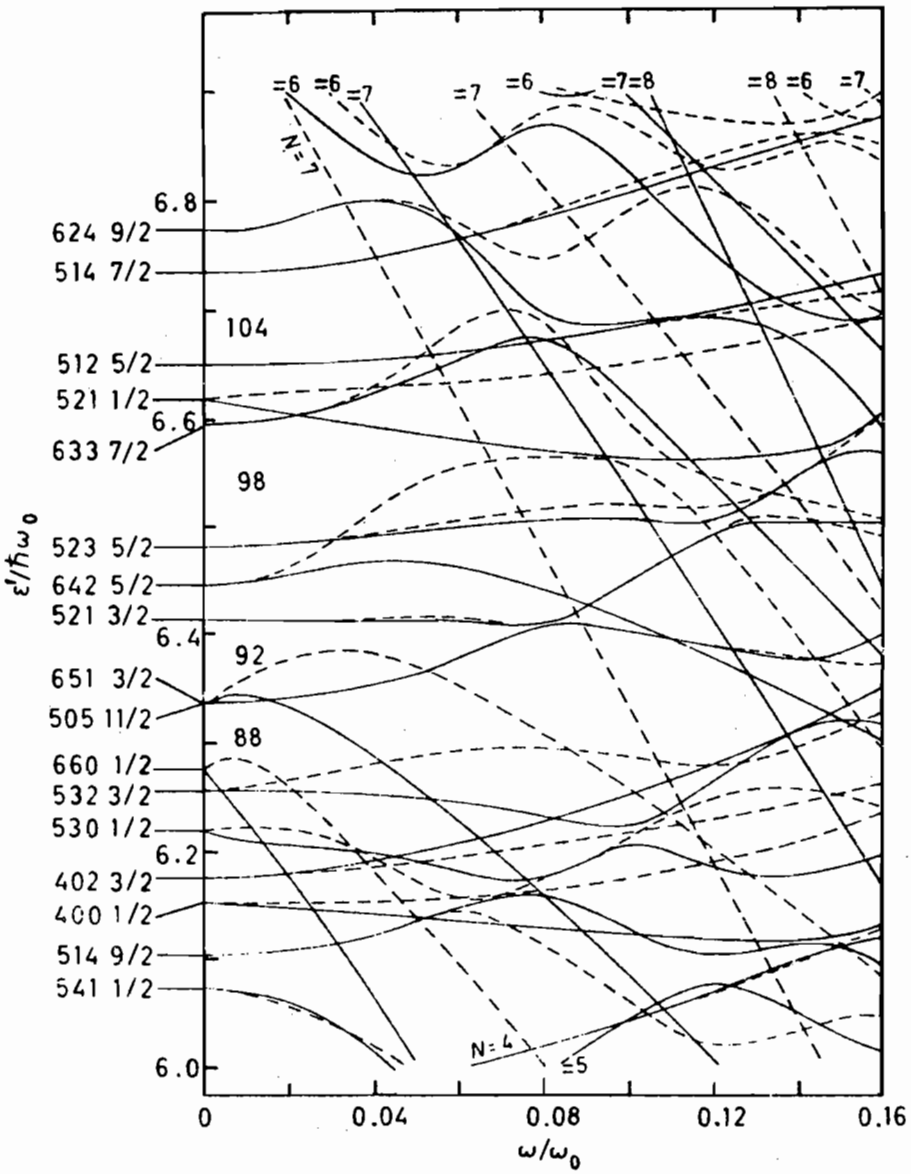


Figure 3.15. Qualitative behavior of the single-particle levels in a cranked Nilsson model at a prolate deformation ($\delta = 0.25$) as a function of the cranking frequency ω in units of $\hbar\omega_0$. Dashed lines correspond to levels with different r_x-quantum numbers. (We are grateful to Dr. R. Bengtsson for the preparation of this figure.)

(vi) Eventually, new shell closures with new magic numbers develop at high angular momenta, which can influence the energy surface for the fission process (see Sec. 3.4.5).

3.4.5 The Deformation Energy Surface at High Angular Momenta

As we already discussed in Section 2.9, the pure Nilsson model cannot be used for the calculation of total energies nor for the calculation of the shape of the energy surfaces at large deformations, because the average part of the energy is not reproduced in a proper way within this model. Therefore, we calculate only the oscillating part of the deformation energy within this model and replace the smooth part by the liquid drop energy at the same deformation.

In the same way, we can calculate energy surfaces at a fixed angular momentum I as a function of the deformation. For an ellipsoidal shape [characterized by the parameters β and γ (Eq. 1.88)], the total energy is then given by

$$E(\beta,\gamma,I)=E_{\mathrm{LDM}}(\beta,\gamma,I)+E_{\mathrm{sh}}(\beta,\gamma,I)-\tilde{E}_{\mathrm{sh}}(\beta,\gamma,I). \quad (3.123)$$

Here E_{LDM} is the deformation energy at a rotating ellipsoid with the rigid-body moment of inertia $\mathcal{I}_{\mathrm{rig}}(\beta,\gamma)$, because one assumes that at high angular momenta pairing correlations can be neglected.* E_{sh} is the shell model energy and is obtained by summing up the single-particle energies. $\tilde{E}_{\mathrm{sh}}$ is the averaged part of it, and is calculated by an appropriate smoothing procedure (see Sec. 2.9).

There are two ways to derive these quantities from the diagonalization of a deformed single-particle potential in the rotating frame: Work either at constant frequency ω [NPF 76] or at constant angular momentum I [ALL 76]. Both methods agree, if one uses a deformed Wood–Saxon potential, where the averaged moment of inertia is very close to the rigid-body value† [BJ 76b, NTP 77].

Several groups have carried out investigations along this line [BLL 75, NP 75, NPF 76, ALL 76, FDG 76, NTP 77] in many regions of the periodic table. Qualitatively they have found similar results:

(i) For spherical or *weakly deformed nuclei* at the beginning of the rare earth region, the nuclei behave similarly to the classical liquid drop (Fig. 1.18): Up to angular momentum $I=50$–$70\hbar$ they are oblate and rotate around the symmetry axis. In this region, the rotation is

*In fact, multiplicity measurement of the γ-cascade indicates the nucleus reaches the rigid-body moment of inertia at high spin values [SBC 76].

†The l^2 term in the Nilsson potential (2.89) is non-local and gives contribution to the effective mass. It produces an averaged moment of inertia which is ~30–40% larger than the rigid body value [Ty 70, 71, BR 71b, Je 73]. One has used scaling procedures to compensate for this effect [NTP 77].

not collective and one expects yrast traps (see Sec. 3.4.7). For higher I-values they rapidly change the shape to triaxial and prolate deformations (Jacobi-shapes; see Sec. 1.7). This transition corresponds to a drastic increase of the angular momentum ("*giant backbending*") as shown in Fig. 3.16. Finally, the nucleus fissions.

(ii) Nuclei in the *middle of the rare earth region* start at low I-values at prolate shapes and rotate around an axis perpendicular to the symmetry axis. With increasing rotation the Coriolis force aligns more and more particles parallel to this axis, and at $I \sim 40$–50 we find a transition to triaxial and sometimes even oblate shapes, * as in Fig. 3.17. At very high angular momenta the nucleus again becomes triaxial and finally fission takes place.

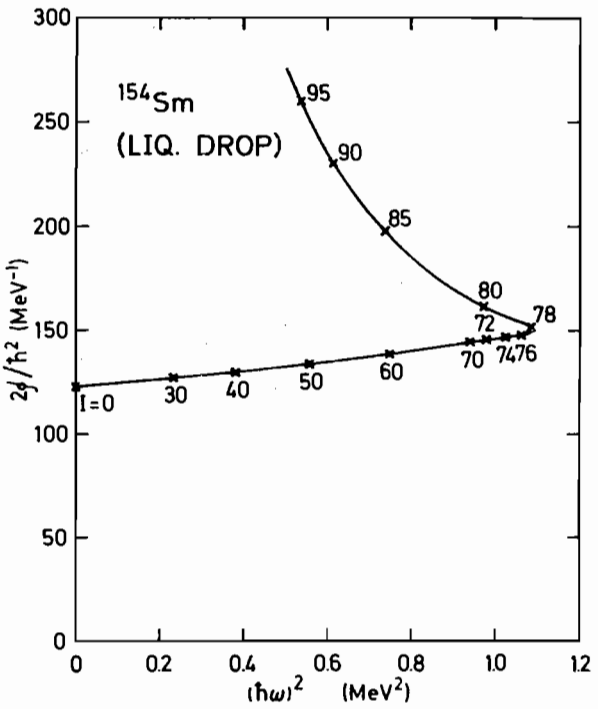


Figure 3.16. Backbending plot for the yrast spectrum of ^{154}Sm in terms of the rotating liquid drop model. (From [ALL 76].)

We want to conclude this section with the remark that such calculations can only give a qualitative impression of the behavior of the nuclear many-body system at such high angular momenta. Only for very low excitations and for energy surfaces with deep minima can we expect the nucleus to have a fixed deformation. In general, it will carry out quantum mechanical zeropoint fluctuations around these minima which ought to be described by a dynamical theory (see, for instance, Chap. 10).

* In calculations based on a Wood–Saxon potential [NTP 77], the nucleus does not reach such drastic γ-deformatione, and fissions without having obtained an oblate shape.

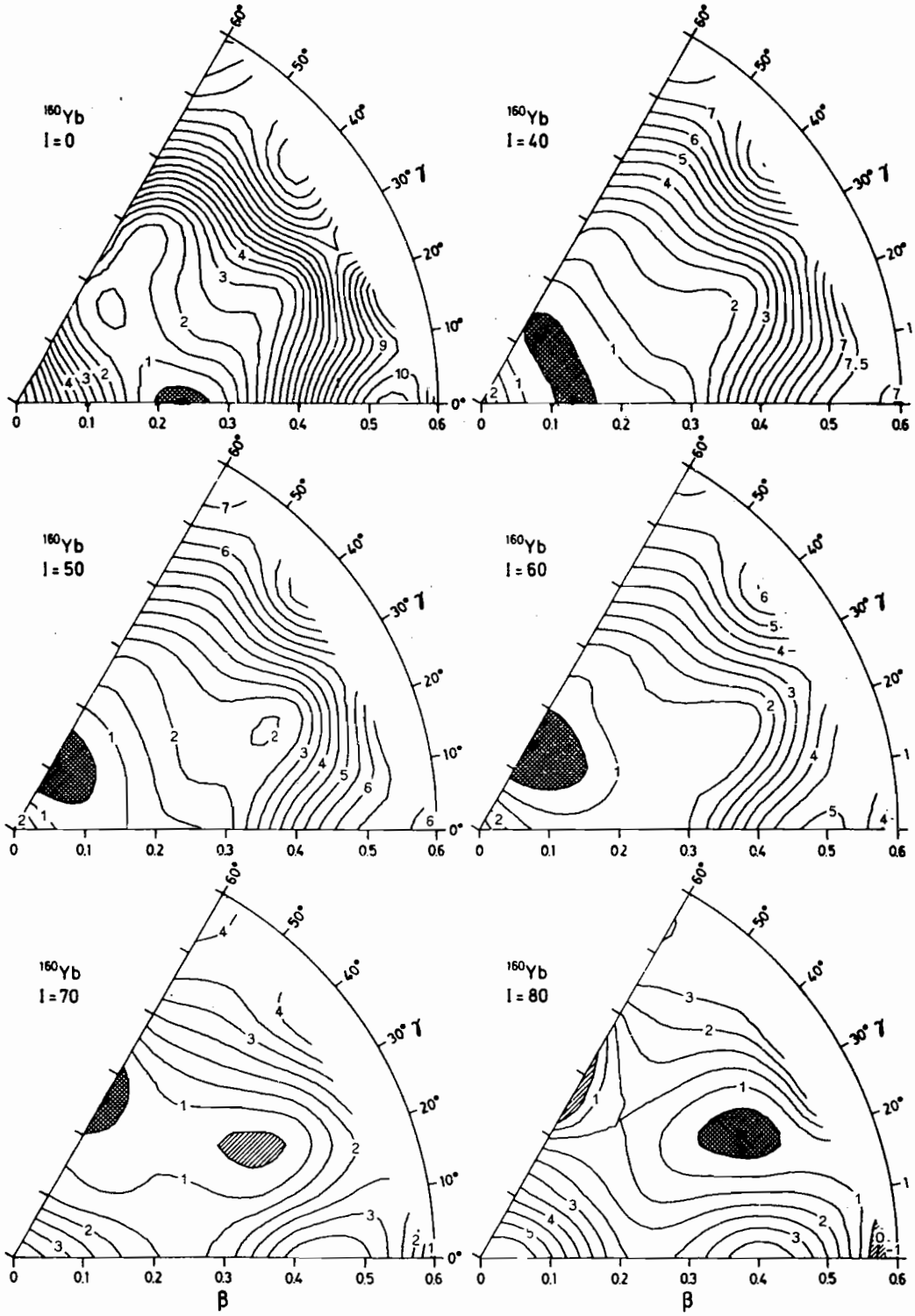


Figure 3.17. Potential energy surfaces in the (β, γ) plane with inclusion of shell corrections for ^{160}Yb as a function of angular momentum. (From [All 76]. Notice that these authors replace the Hill–Wheeler coordinate γ by $-\gamma$.)

3.4.6 Rotation about a Symmetry Axis

As we have seen in the last sections, there are regions in the periodic table where we find nuclei "rotating" around the symmetry axis. This happens in particular at the beginning and end of each major shell. At the beginning of each shell are a few particles sitting on a more or less spherical core. To create a large amount of angular momentum they all must align along the rotational axis (the x-axis), that is, classically speaking these particles have to run in the equatorial plane around the nucleus and produce an oblate density distribution.*

Such a configuration is certainly not collective rotation, since there are only a few particles involved. Each single particle wavefunction is an eigenfunction of j_x with the eigenvalue α_i. The component of the total angular momentum in the direction of the symmetry axis is given by

$$\langle J_x \rangle = \sum_{i=1}^{A} \alpha_i \tag{3.124}$$

To increase the angular momentum we have to change the occupation in the deformed well and to put particles from levels with lower (for instance negative) α-values into those with higher (positive) α-values.

Formally, this can be done again by a cranking procedure around the symmetry axis. Since the operator j_x commutes with h, we get the *single-particle energies in the "rotating frame."*

$$\epsilon_i' = \epsilon_i - \omega\alpha_i, \tag{3.125}$$

where ϵ_i are the eigenvalues of h in the nonrotating frame. These are straight lines as a function of ω (Fig. 3.18a) whose slope is given by

$$\frac{d\epsilon_i'}{d\omega} = -\alpha_i. \tag{3.126}$$

The condition to minimize the energy in the rotating frame

$$E' = \sum_{i=1}^{A} (\epsilon_i - \omega\alpha_i) \tag{3.127}$$

guarantees that one always occupies the lowest levels ϵ_i'. With increasing frequency ω we thus obtain a stepwise increasing of the angular momentum (Fig. 3.18b). The distance between two steps and the size of the steps is given by the distances of the levels ϵ_i and the angular momentum values α_i. Therefore there is a statistical increase of the angular momentum with the frequency ω. The *moment of inertia* $\mathcal{I}$ is defined only on the average (dashed line in Fig. 3.18b).

To get an estimate for the size of this moment of inertia [Bo 76b], we realize that each line in Fig. 3.18a has the slope $-\alpha_i$ (increasing or

* At the end of a major shell similar arguments can be applied for holes. We then end up with a rotation about the symmetry axis of a prolate density distribution. In the middle of a shell there are many valence particles forming a prolate deformed shape. It is easy to generate large angular momentum from a few alignment processes, without too drastic changes in deformation.

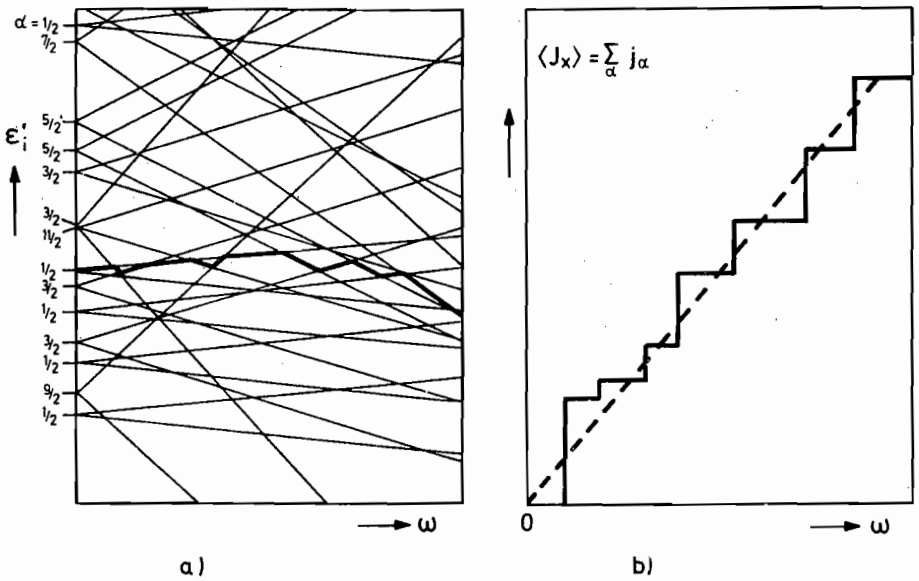


Figure 3.18. (a) Schematic representation of the eigenvalues ϵ_i' in Eq. (3.125) as a function of the cranking frequency. (b) The angular momentum $\langle J_x \rangle$ obtained by increasing values of ω and by always occupying the lowest levels in (a).

decreasing, depending on the sign of α). For $\omega = 0$ the levels $\pm\alpha_i$ are all occupied pairwise. The resultant angular momentum vanishes. For finite ω some of these levels with negative α are no longer occupied, but other downward-coming levels with positive α-values are occupied anew. For a fixed value of α (for instance $\alpha = \frac{1}{2}$) the number of newly occupied levels is on the average $\omega\alpha g(\alpha)$, where $g(\alpha)$ is the density of levels with the quantum number α. Therefore, we obtain for the angular momentum on the average

$$\langle J_x \rangle_{\text{av}} = \sum_{i < \epsilon_F} \alpha_i = \sum_{\alpha} \alpha \cdot \omega \cdot \alpha \cdot g(\alpha) = \omega \sum_{\alpha} \alpha^2 \cdot g(\alpha), \tag{3.128}$$

and for the average moment of inertia

$$\mathcal{I}_{\text{av}} = \sum_{\alpha} \alpha^2 g(\alpha). \tag{3.129}$$

An evaluation of this quantity within the Thomas–Fermi approximation gives exactly the rigid-body value for the moment of inertia [Bo 76b]. On the average, "single-particle" rotation around the symmetry axis therefore shows quite a similar behavior to collective rotation, although the internal structure is completely different.

3.4.7 Yrast Traps

Since the level spacing in the case of a rotation around the symmetry axis has a statistical character, one expects *yrast traps* (high spin isomeric states). They are defined as yrast states which cannot undergo a rapid γ-transition and have been predicted by Bohr and Mottelson [BM 74].

Filling the levels in the rotating well always from the bottom, we eventually obtain, with increasing angular velocity, jumps in the angular momentum by several units (see Fig. 3.18b). This can be visualized most easily in a representation of the eigenvalues ϵ_i in the nonrotating oblate deformed Nilsson well as a function of the components α of the angular momentum along the symmetry axis (Fig. 3.19). For small deformations we have rather pure j-configurations. The eigenvalues ϵ_i for levels in the same j-shell lie on approximate parabolas [see Eq. (3.39); we have only to replace Ω by α]. At oblate deformations ($\beta<0$) the highest α-values, that is, the most aligned states, have the deepest energy, because their oblate density distribution in the equatorial plane has the maximal overlap with the oblate density of the core.

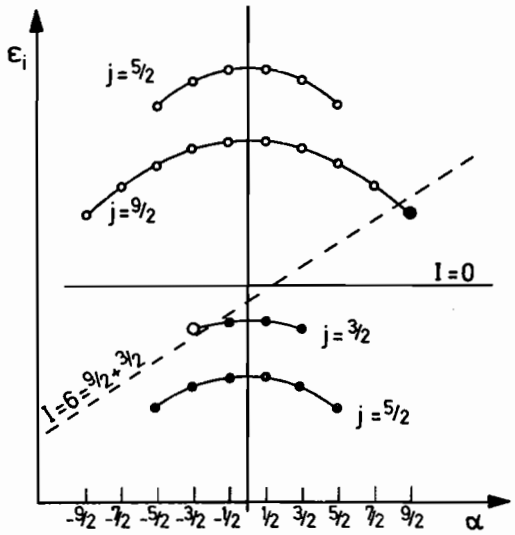


Figure 3.19. Schematic representation of the single-particle energies ϵ_i in an oblate deformed Nilsson well as a function of α (the eigenvalue of j_x) with the Fermi surface at $I=0$ (full line) and at $I=6$ (dashed line).

At $\omega=0$, the Fermi surface is a horizontal line and all levels below it are occupied. At finite ω-values, according to Eq. (3.125), the Fermi level in the rotational frame has a slope ω. As the value of ω increases, levels with negative α-values are vacated and those with positive α-values are newly occupied. In Fig. 3.19 we show a situation where a jump of 6 units of angular momentum occurs. The remaining I-values on the yrast line can be obtained by constructing *ph*-configurations with respect to the rotated energy surface. This is usually connected with an increase in energy.

Therefore, it may happen that the yrast line is no longer a monotonically increasing function of I, and a certain level with $I=I_0$ can have a deeper energy than the neighboring levels with $I=I_0-1, I_0-2, \ldots$. A fast γ transition of $E1$-, $E2$-, $M1$- or $M2$-character (i.e., with low multipolarity; see Appendix B) is then not allowed (so-called *energy spin traps*).

Even if this is not the case, the internal structure of the states with I_0-1, I_0-2 might be rather different from the structure of the state I_0. Then the transition matrix elements can be very small and the lifetime of the state I_0 is very large (*structure spin traps*). Examples are cases where the states with I_0-1 or I_0-2 can only be reached from the state I_0 by a $2p$-$2h$ excitation, or cases where the γ-transition matrix element is hindered by the intrinsic selection rules of the single-particle wave functions (for instance, l-forbidden $M1$-transitions; see Sec. 2.7.2.).

Several groups have investigated the theoretical possibility of yrast traps by searching for cases of rotation around the symmetry axis in the energy surfaces at high angular momenta and by investigating the detailed single-particle structure in these cases [ALL 76, CDS 77, AK 77, DNM 77, PFL 78, PTF 78, AHL 78, Ab 78, MDN 78]. Some regions in the periodic table have been found in which they should be expected. These are mostly weakly deformed nuclei such as Te, Ba, Ce, Sm, neutron deficient rare earth isotopes, and nuclei in the Pb-region.

On the *experimental side* we have known for a long time about states with large angular momenta and very long lifetimes for spherical nuclei. These are states with rather pure high j-shell configurations, which have no allowed γ-transition matrix elements of low multipolarity to other states at deeper energy. The most famous example [PAG 62] is the 18^+ isomer in ^{212}Po with a lifetime of 45 sec, which consists of an aligned $(\pi h_{9/2})^2(\nu i_{11/2})^2$ configuration.

Another group of high spin isomers, usually called *K-isomers*, were observed in well deformed nuclei, such as the $4s$ and the $31y$ isomers in ^{178}Hf, which can be interpreted as a $K^\pi=8^-$ and a $K^\pi=16^+$ band head, whose γ-decay is K-forbidden [HR 68, KL 77].

Finally, in recent years an island of about 20 adjacent nuclei in the light rare earth region with long lived isomers has been found [PBB 77]. They are probably associated with oblate deformation and "single-particle" rotation [AHL 78].

The properties of nuclei in the vicinity of ^{208}Pb have usually been studied in terms of the spherical shell model with residual interaction and a proper angular momentum coupling. Since these configurations are rather pure, some calculations of this type could explain measured excitation energies with extraordinary accuracy [BBH 77].

The physical reason for the aligned high spin configurations having such a long lifetime is the same in this picture as we have explained in the model of rotations around a symmetry axis: Two nucleons with aligned spins gain energy because their residual interaction is largest for wave functions with a large spacial overlap* (see Sec. 4.4.8.). Therefore, these high spin levels are lower in energy than the neighboring states with smaller I-values, and fast γ-transitions are forbidden. In the description of

*This effect has been called the MONA-effect (Maximal Overlap of the Nuclear wave function by Alignment [FPD 76]).

these nuclei by the mean field approach in a rotation about a symmetry axis, the same effect shows up in the fact that (as discussed in Fig. 3.18) aligned levels with maximal α-values and oblate density distribution are shifted downwards in energy because of their large overlap with the oblate core.

Both methods—shell model calculations in a spherical basis based on angular momentum coupling techniques and cranking calculations in a slightly deformed well—are certainly hard to compare. The spherical shell model is certainly much better (though it takes more effort) as long as one can assume the spherical core to be inert, otherwise one has to take into account multiparticle–multihole configurations. It is therefore limited to the very close vicinity of magic nuclei. On the other side, the cranking model treats the residual interaction by a deformed well and is therefore much easier to handle over a wide range of nuclei. It allows, however, only very qualitative predictions, since it violates conservation of angular momentum and is microscopically derived only in the regions of well deformed nuclei with rotations perpendicular to the symmetry axis (see Sec. 11.4).